HORNET
NAVAL STRIKE FIGHTER



Spectrum HoloByte

FLIGHT MANUAL

A WORD ABOUT THIS GAME

We would like to thank you for purchasing one of our products. For your convenience you can make backups of the original disk(s) in order to protect your investment. Please remember that the developers of this product are relying on you not to give away copies to others. They have spent a lot of time, effort and money to bring this software to you, and are depending solely upon royalties from sales in order to make a living. Your purchase will allow them to invest the time and energy necessary to produce additional high quality software in the future.

We are committed to providing products for you that are of the highest quality and value. If you have questions concerning any of our software, or if there are additional programs that you would like to see developed, please contact us at:

Spectrum HoloByte
2490 Mariner Square Loop
Alameda, CA 94501
(510)522-1164
(510)522-3587 FAX

America Online: S HOLOBYTE
CompuServe: 76004,2144
GEnie: HOLOBYTE
Internet: 76004.2144@compuserve.com
Prodigy: TXHJ33A



Gilman G. Louie
Spectrum HoloByte, Inc.

HORNET
NAVAL STRIKE FIGHTER



FLIGHT MANUAL

Spectrum HoloByte

Hornet credits

Based on the Original Design Concept of *Falcon*
Gilman "Chopstick" Louie

Software Engineers
Michael "Papa Bear" Mancuso
Gary "RC" Stottlemyer
Dan "Bullfrog" Deremiah
Heather "Tigger" Mace
Kuswara "Porkchop" Pranawahadi

3-D World Builders
Anthony "Sterculius" Chiang
Lawrence "Trim" Kevin
Eric "Alien" Grotke

Artists
Jim "2-5 India" Voytilla
Lawrence "Trim" Kevin

Product Manager
Steve "Grout" Blankenship

Assistant Product Manager
Robert "Jocko" Cieri

Product Marketing Manager
Neil "Waxer" Johnston

Quality Assurance
Evan "Baron" Birkby
Jack "Lumber" Gibbons
Gabe "Appolo" Turk
Derwin "Panther" Wyatt
Kurt "Goat Killer" Boutin

Music Composer
Paul "Moggy" Mogg

Music Arrangers
Andy "Check Mix" Edlen
J. "Fretless" White

Research and Design Specifications
Robert "Jocko" Cieri

User Manual Writing
Geoff "Too Cool" Gilpin
Lisa "Minerva" Ruffolo
Kyle "Bulldog" Brink
Robert "Jocko" Cieri

User Manual Editing
Marisa "No. 19" Ong
Robert "Spaceman" Giedt
"Smokin'" Joe Ward

Illustrator
Lawrence "Trim" Kevin

Manual Design
Carrie "Ethyl" Galbraith

Photos Courtesy Of
The Department of Defense
McDonnell Douglas
Hughes Missile Systems
Darryl Shaw

Special Thanks To
Robert "Mouse" Shaw
Jack "C.J." Connell
Lt. Bill "Hack" McMasters, USN
Dave "Nauga" Hyde
Al Bowers, Ames/Dryden FRF
Daryl L. Stephenson
Darryl Shaw
The Software Resource
The 510th TFW
Julie Bailey at C.J. Krehbiel
Kathryn Lynch
Anne Berg

If you have any questions about *Hornet: Naval Strike Fighter* or any of our other products, please contact Spectrum HoloByte Customer Support at:

✍ Spectrum HoloByte
2490 Mariner Square Loop
Alameda, CA 94501
ATTN: Customer Support

☎ (510)522-1164
9:00 AM to 5:00 PM Pacific Time
Monday through Friday

✆ Fax (510)522-3587

America Online:

To reach our Customer Support board in the Industry Connection, press Ctrl K for "Go to Keyword." Then type **SPECTRUM** in the Keyword window. In addition to posting and reading messages, you can download files (demos, updates, troubleshooting tips, etc.) from the "Software Libraries." You can also send electronic mail to Customer Support at S HOLOBYTE.

CompuServe:

To reach our Customer Support board in the Game Publishers B Forum, type **GO GAMBPUB** at any "!" prompt. Then select "Section 3" for Spectrum HoloByte. In addition to posting and reading messages, you can download files (demos, updates, troubleshooting tips, etc.) from the "Libraries (Files)" menu. If you're looking for head-to-head opponents, look for other *Hornet: Naval Strike Fighter* players in the Modem Games Forum by typing **GO MODEMGAMES**. You can send electronic mail to Customer Support at 76004,2144.

If you are not already a member of CompuServe, you can call CompuServe toll-free at 1-800-848-8199 and ask Representative #142 for your free introductory membership and $15 usage credit. Besides online support of *Hornet: Naval Strike Fighter* and our other Spectrum HoloByte products, CompuServe offers many other services including communications, hardware/software support, travel, reference libraries and more.

GEnie:

To reach our Customer Support board in the Games RoundTable, type **M805;1** at any "?" prompt. Then select "Category 18" for

Spectrum HoloByte. In addition to posting and reading messages, you can download files (demos, updates, troubleshooting tips, etc.) from the "Games RoundTable Libraries." You can also send electronic mail to Customer Support at HOLOBYTE.

Internet:

You can send electronic mail to Customer Support at 76004.2144@compuserve.com.

Prodigy:

You can post and read messages in the "Flight Simulators" topic on the Game Club bulletin board (located in the Game Center area). You can also send electronic mail to Customer Support at TKNI33A.

Spectrum HoloByte Customer Support BBS:

To reach our Customer Support BBS, phone (510)522-8909 [8-N-1]. The BBS has four lines and is open 24 hours a day. The BBS supports from 300 baud to 14.4K (v.32bis) and Xmodem, Ymodem and Zmodem transfer protocols. After you are logged in, just follow the onscreen prompts to sign on as a new user. The Customer Support BBS has message areas and file libraries (program updaters and game utilities).

If you are having problems with *Hornet: Naval Strike Fighter* we can best help you if (1) you are at your computer when you call, and (2) you have the following information handy:

- Version number or serial number of *Hornet: Naval Strike Fighter*
- Your computer's brand and model
- Your computer's BIOS brand and version number
- Total RAM installed in your computer
- Name and version number of your operating system (MS-DOS or DR DOS)
- Name and version number of your memory manager (such as QEMM or 386Max)
- Video card brand and model name
- Mouse brand and version number of mouse driver
- Sound card brand and model name
- Contents of AUTOEXEC.BAT and CONFIG.SYS files
- Type `MEM /C` at the DOS prompt and copy down the onscreen listing

Hornet and MiG-29 © 1993 Spectrum HoloByte, Inc. Falcon 3.0 © 1991–93 Spectrum HoloByte, Inc. All Rights Reserved. MiG-29, MiG-29: Deadly Adversary of Falcon 3.0, Falcon 3.0 and Operation: Fighting Tiger are trademarks, and Falcon and Spectrum HoloByte are registered trademarks of Spectrum HoloByte, Inc. Head to Head® is a registered trademark of Hasbro, Inc. and is used under license. Other trademarks are the property of their respective holders.

Printed in the United States of America

First Edition, November 1993

10 9 8 7 6 5 4 3 2 1

Table of Contents

1 Introduction 1
In this package 2
Before you start 2
System requirements for *Hornet* 3
Installing *Hornet* 4
If you want to install *Operation: Fighting Tiger* 4
Starting the Install program 5
Viewing system requirements 6
Installing *Hornet* on a hard drive 6
Starting *Hornet* 7
About the Hornet 8

2 Cockpit 11
Cockpit views 12
Cockpit look-up view 13
Stores control panel 13
Launch light 13
Lock light 14
Warning light 14
Caution light 14
Flare/chaff indicators 14
Stall light 14
Auto light 14
Brks light 14
Disc light 14
Gear light 14
Hook light 15
Caution light cue panel 15
Angle of attack indexer 15
ECM light 15
Flps light 15
IFF light 15
G-limiter light 15
AOA limiter 16
Cockpit look-down view 16
Standby radar warning receiver (RWR) 17
Standby airspeed indicator 17

Standby fuel indicator....17
Standby attitude display indicator (ADI)....17
Standby barometric altimeter....17
Standby rate of climb indicator....17
Left view....18
Right view....18
FLAP caution light....19
BURN caution light....19
RADAR caution light....19
FUEL caution light....19
HUD caution light....20
NWS caution light....20
STORE caution light....20
OX LO caution light....20
ECM caution light....20
ENG caution light....20
JAM caution light....21
NAV caution light....21
BRAKE caution light....21
ARM caution light....21
WING caution light....21

3 Multifunction Display Group....23
Digital display indicators....24
Engine monitor....25
Radar....27
Maverick....32
Radar warning receiver....32
Attitude display indicator....33
Horizontal indicator....34
HSI (horizontal situation indicator)....35
Map mode....39
Combined HSI and navigational map mode....40

4 Head-Up Display....41
HUD basics....42
Heading scale....43
Waypoint symbol....43

Airspeed 43
Altitude 43
Angle of attack 43
Mach number 43
G force 43
Maximum G 44
Velocity vector marker 44
Vertical velocity 44
Pitch ladder 44
Waypoint distance 44
RPM AB indicators 44
Air-to-air HUD mode 45
HUD for A-A missiles 45
HUD for A-A cannon 49
Air-to-ground HUD mode 51
Mode 51
Weapon type and quantity 51
True airspeed 52
Computer-controlled release point (CCRP) 52
Displayed impact line 52
Instrument landing system 53
Horizon bar 54
Angle of attack bracket 54
Bank angle scale 54

5 Air Combat Training 55
Lesson 1: Takeoff and landing 57
Take off 57
Circle the airfield 57
Landing 58
Lesson 2: Air-to-air combat 60
Lesson 3: Air-to-ground combat (smart weapons) 62
Lesson 4: Air-to-ground combat (bombing) 63

6 Carrier Operations 67
Carrier take-offs and landings 68
Takeoff 69
Landing 70

7 Communications **81**
Falcon 3.0 and *MiG-29* compatibility 82
Updating *Falcon 3.0* and *MiG-29* 83
The Caller 84
Head-to-head game formats 84
Duels 85
Single game duels 85
Multigame duels 85
Setting up a multiplayer multigame duel 86
Campaigns 88

8 Theater Briefing **89**
Bosnia-Hercegovina: The enigma and the terror 90
Campaign rationale 93
Campaign objectives 94

Appendix A. Technical Specifications **98**

Appendix B: Naval Medals **102**

Appendix C: Glossary **104**

Appendix D: Bibliography **108**

CHAPTER

1

INTRODUCTION



The mission of the "Hornet" aircraft is to keep the peace in the air and on the ground. If necessary, Hornet pilots can clear the air of enemy fighters. The main purpose of the F/A-18 Hornet, however, is to attack ground targets.

The Hornet shares many features with its cousin, the F-16 "Falcon." Both are multipurpose, "fly-by-wire" aircraft that serve in the air forces of several NATO countries. Both planes use a similar, and very large, arsenal of air-to-air and air-to-ground weapons.

One feature that distinguishes the Hornet from the F-16 is its base of operations. The Hornet flies from an aircraft carrier. Also, it has some unique instruments to suit its air-to-ground mission.

Hornet: Naval Strike Fighter puts you in the cockpit of this right maneuverable aircraft. *Hornet* is a companion to your copy of *Falcon 3.0* or *MiG-29*. When you install *Hornet*, you get a new aircraft, a new theater of operations and additional features, including an aircraft carrier.

In this package

Your *Hornet* package should contain:

- Three 1.44MB 3½" disks
- *Hornet Flight Manual*
- *Hornet Briefing for MiG-29 Pilots*
- *Hornet* Quick Reference Card
- Theater map of Bosnia-Hercegovina
- Registration card

Before you start

Have you already installed *Falcon 3.0* or *MiG-29*? If not, don't go any further. Your computer must have a working copy of *Falcon 3.0* or the stand-alone version of *MiG-29* in order to run *Hornet*.

Hornet has the same basic hardware and software requirements as *Falcon 3.0* or *MiG-29*. If your system

can run *Falcon 3.0* or *MiG-29*, all it needs to run *Hornet* is a little more space on your hard drive. For detailed requirements, see the next section.

System requirements for *Hornet*

Hornet requires:

- *Falcon 3.0* or *MiG-29* installed on your hard drive
- 12MHz 80286 compatible computer
- 1MB RAM minimum
- 603K (618,472 bytes) of free conventional memory
- DOS 5.0 or DR DOS
- VGA graphics
- One 1.44MB 3½" disk drive
- Hard drive with 7MB free

We recommend the following peripherals to enhance game play:

- Joystick, dual joysticks, CH FlightStick Pro, ThrustMaster Flight Control System (FCS) or rudder pedals
- ThrustMaster Weapons Control System (WCS)
- Mouse
- AdLib, Sound Blaster, Roland MT-32 or Roland LAPC-1 sound card
- 2MB RAM with EMS (expanded memory)
- Expanded memory manager such as EMM386, QEMM or 386Max
- Null modem serial cable or 2400 baud modem for two-player communications
- Novell NetWare or Novell NetWare Lite network for multiplayer communications

For optimum performance, you should have the following:

- *Falcon 3.0* or *MiG-29* installed on your computer

- *Operation: Fighting Tiger* installed on your computer
- 33MHz or faster 80486 compatible computer
- 4MB RAM with EMS (expanded memory)
- 603K (617,472 bytes) of free conventional memory
- DOS 5.0, DOS 6.0 or DR DOS 6.0
- 16-bit VGA graphics
- One 1.44MB 3½" disk drive
- Hard drive with 7MB free
- Dual joysticks, CH Flight Stick Pro with rudder pedals, or ThrustMaster FCS and WCS with rudder pedals
- Sound Blaster and either a Roland MT-32 or Roland LAPC-1 sound card

Installing *Hornet*

Your copy of *Hornet* comes with an installation program that automatically sets up the game on your hard drive. During the installation process, *Hornet* requires 7MB of free space on the hard drive that contains *Falcon 3.0* or *MiG-29*. Once it's installed, *Hornet* only needs 5.5MB.

Installing *Hornet* will update your copy of *Falcon 3.0* to version 3.03 and your copy of *MiG-29* to version 1.2. These updated versions of *Falcon 3.0* and *MiG-29* will be communications compatible with *Hornet*. Please see Chapter 7: Communications for more information.

If you want to install *Operation: Fighting Tiger*

If you have *Falcon 3.0* and *Operation: Fighting Tiger* you must install in the following order:

1. *Falcon 3.0*
2. *Operation: Fighting Tiger*
3. *MiG-29: Deadly Adversary of Falcon 3.0* (the add-on version)
4. *Hornet*

If you have the stand-alone version of *MiG-29* and *Operation Fighting Tiger*, you must install in the following order:

1. *MiG-29*
2. *Operation: Fighting Tiger*
3. *Hornet*

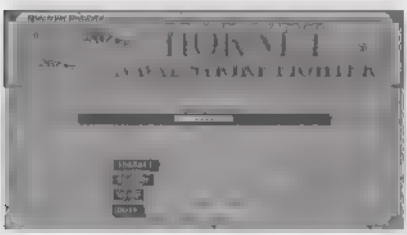
Do not attempt to install *Operation: Fighting Tiger* after you install *Hornet*. If you have already installed *Hornet* and wish to add *Operation: Fighting Tiger*, you must re-install everything in the above order.

When re-installing, you must delete all the old files and subdirectories in your FALCON3 or MiG29 directory before you begin the re-installation.

Starting the Install program

To install *Hornet* on your hard drive, start your computer and follow the procedure below:

1. Insert *Hornet* Disk 1 into drive A or B.
2. From the DOS prompt, type **A:** or **B:** as appropriate and press Enter.
3. Type **INSTALL** and then press Enter. The *Hornet* installation procedure begins. Here's the screen you should see:



In general, you use the Install program by selecting items on the screen or pressing buttons. You can do other operations using the mouse or keyboard. For instance, on the main screen above, you can choose one of the

menu items by pressing the → key until it is highlighted. Then press Enter.

To select a menu item with the mouse, move the mouse pointer over the line and double-click the mouse button. That is, press it twice in quick succession. You can also click the item once and then click the Okay button.

The Install program has a number of options besides installing the software. Before you begin the installation process, you may want to take a look at the Read Me file, a text file containing important, up-to-the-minute information. Select the View Readme menu option to see this document.

Many options in the Install program take you through a series of tasks and screens. Many screens have a Go Back button that takes you to the previous screen. You can press the Esc key for the same effect.

Viewing system requirements

If you're in doubt about the ability of your system to run *Hornet*, select the View Requirements option. You will then see a list of hardware and software items.

If any items on the list appear highlighted in red, you won't be able to install the software without making changes to your system. If an item appears in yellow, you can proceed with installation, but *Hornet* may not work until you make the changes.

In many cases, the Install program can suggest a change to fix a highlighted problem item. Click the Suggest Fix button to view the suggestion. It may advise you to make changes to your system startup files, add RAM, start over with fewer TSRs. These and other changes, like deleting excess files on your hard drive, have to be done outside the Install program in DOS.

Installing *Hornet* on a hard drive

The following procedure copies the *Hornet* files from the floppy disks to your hard drive.

1. If you haven't already done so, run the Install program as described in the section *Starting the install program*.

3. Select the Install Hornet option. The window that appears shows a list of system requirements.

4. Since we're assuming that your system is ready for *Hornet*, click Okay to continue with the installation. The install program will next search for *Falcon 3.0* or *MiG-29*.

4. Choose the name of a subdirectory to contain *Hornet*. To accept the default name, just click Okay.

   Click Continue if you want to install Hornet in another directory, then enter the name of the directory you want to install Hornet to.

5. You will then see a summary screen showing the source and destination drives and subdirectories. The destination directory must contain an installed version of *Falcon 3.0* or *MiG-29*.

   Click Install. When the Install program finishes with one disk, it stops and asks you to swap disks. Remove the one currently in the drive, replace it with the next one in the series and click Okay.

Starting *Hornet*

Once you've installed *Hornet*, you're ready to fly! To start the game, follow these steps:

1. Get to the disk drive and subdirectory containing *Hornet*. For example, if *Hornet* was installed to D:\FALCON3, type the following at the DOS prompt.

   D: Enter
   CD \FALCON3 Enter

2. To run *Hornet*, type:

   HORNET Enter

After an animated opening sequence and the title screen, the next thing you see is the War Room.

Once you get to the War Room, you should know what to do based on your experience with *Falcon 3.0* or *MiG-29*. The rest of this manual covers the unique features of *Hornet*. Most of these have to do with the airplane itself and the new theater of operations, Bosnia.

Herzegovina. If you have questions about a feature of the game that's doesn't appear in this manual, consult your *Falcon 3.0 Flight Manual* or *MiG-29 Reference Manual*.

Note: Included in the *Hornet* package is a special report for MiG-29 owners entitled *Hornet Briefing for MiG-29 Pilots*. This document will cover any differences between MiG-29 and Hornet that are not covered in either the *Hornet Flight Manual* or the *MiG-29 Reference Manual*.

About the Hornet

The McDonnell Douglas F/A-18 "Hornet" is a single or two seat, twin engine strike fighter flown by the U.S. Navy and Marine Corps and the air forces of Canada, Australia, Spain and Kuwait. Designed for air-to-air and air-to-ground missions, it can perform as an effective air superiority fighter, escort, air defense, suppression of enemy air support, and day/night strike. Its all-weather performance, coupled with Navy and Marine Corps missions and to employ precision guided weapons gives tactical planners unprecedented flexibility in mission planning.

Originally a Northrop design, the prototype Hornet was designated the YF-17 and competed with the YF-16 in the Light Weight Fighter (LWF) and Air Combat Fighter (ACF) programs. The LWF program was a research exercise begun in 1972 to test the viability of cheaper, lighter and more agile fighter designs. The program was upgraded to the ACF program after development began when pressure from the Eastern bloc countries required a new generation of fighter aircraft.

After extensive competitive trials in 1974 the Navy chose the YF-17. The Air Force chose the YF-16, which became the F-16 Fighting Falcon. After further development the YF-17 became the F/A-18 Hornet.

While McDonnell Douglas builds the F/A-18 as a whole, the airframe is assembled by Northrop. The Hornet rides two General Electric F404 smokeless afterburning turbofans and Hughes provides the look-down, shoot-down radar.

Introduction

The Hornet first flew on November 18, 1978. It entered operational service with the Navy and Marine Corps in 1983 as a replacement for the A-7 light attack aircraft and the F-4 Phantom II fighter. Today the Navy and Marine Corps have nearly 850 Hornets in 49 squadrons. More than 1,280 F/A-18s have been delivered worldwide. The Navy operates 30 squadrons from bases in Florida and California, while the Marine Corps operates 19 squadrons from bases in California, South Carolina and Hawaii.

The F/A-18 was designed to be both reliable and maintainable, ready for duty ashore or at sea. In Navy and Marine Corps operations, the F/A-18 is demonstrating three times better mission availability at half the maintenance of other fleet aircraft.

Since its introduction, the Hornet has been continually improved. In 1987, F/A-18C and D models introduced new self-defense capabilities, including the ability to carry the Advanced Medium Range Air-to-Air Missile (AMRAAM) and the infrared imaging Maverick air-to-ground missile.

Standard air-to-air combat armament for today's F/A-18 includes two heat-seeking AIM-9 Sidewinder missiles on the wingtips, two radar-guided AIM-120 AMRAAMs on the fuselage stations, and an internally mounted M61A1 20mm gun with 570 rounds of ammunition.

Air-to-ground weapons can be carried on centerline, inboard or outboard stations. Up to 15,500 lbs. of payload can be carried on nine external stores stations. Ferry range exceeds 1,800 nm.

The Hornet's first combat action was in Libya in 1986. Navy and Marine Corps F/A-18s deployed on board the USS Coral Sea flew combat air patrol missions and knocked out a Libyan surface-to-air missile site at Sirte.

During Operation Desert Storm in 1990-1991, more than 200 U.S. and Canadian Hornets flew continuous missions in support of the Coalition's drive to liberate Kuwait from occupation by Iraq. F/A-18s participated in virtually every conceivable mission area and shot

down two Iraqi fighters. U.S. Hornets flew more than 11,000 sorties and recorded availability rates of 99%.

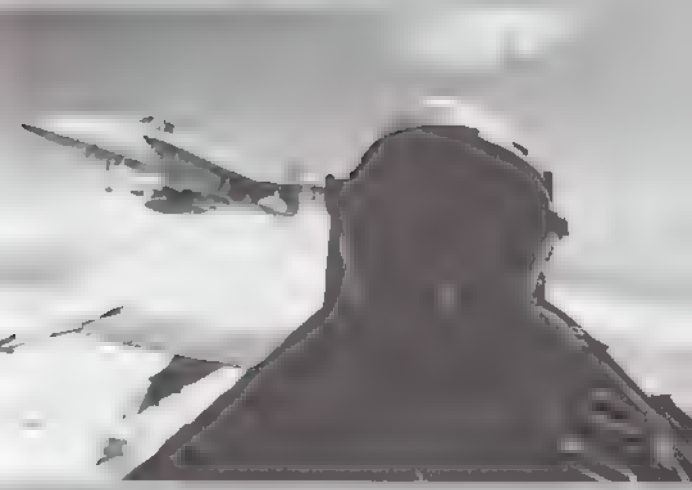
The Hornet's ability to convert rapidly between fighter and attack modes was demonstrated on January 17, 1991, when aerial encounter between two F/A-18s from the USS Saratoga and two Iraqi MiG-25s. The F/A-18s were loaded for an attack mission with 8,000-lbs of bombs. The Hornet pilots switched to high-altitude mode shot down the MiGs, switched back to attack mode hit the assigned ground targets, then returned safely to the Saratoga.

F/A-18s were deployed in the Persian Gulf region again in 1992 as part of Operation Southern Watch to enforce the Coalition's no-fly zone in Iraq. Hornets from the aircraft carrier USS Kitty Hawk flew combat air patrol and suppression of enemy air defense missions over southern Iraq in January 1993. In December 1992, F/A-18s from the Kitty Hawk supported U.S. relief efforts in Somalia as part of Operation Restore Hope.

In *Hornet*, you will be flying the F/A-18 in combat in any of the theaters from *Falcon 3*, *MiG-29* or *Operation Fighting Tiger* if you have that game, plus a seventh theater straight from today's headlines: Bosnia Herzegovina. Flying missions from your home on the carrier USS Theodore Roosevelt, you will experience what today's Navy and Marine Corps pilots must face as the front line members of the United States' global peacekeeping effort.

CHAPTER
2

COCKPIT

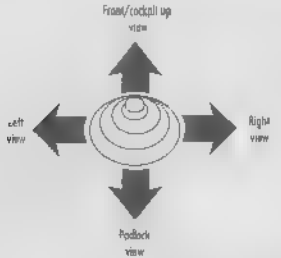


As a Hornet pilot you rely on many of the instruments, controls and weapons you learned to use while flying the F-16 in *Falcon 3.0*. For the Hornet in *MiG-29*, some of these appear unchanged in the Hornet cockpit. For example, the stick and throttle work the same way.

Other items offer similar functions in a slightly different form. The airspeed indicator in the Falcon and Fulcrum HUDs is an analog scale, while a digital display in the Hornet. Finally, some Hornet features, such as the configurable cockpit displays, are unique.

Cockpit views

In *Hornet*, you can view game action from inside or outside your plane as you do in *Falcon 3.0* or *MiG-29*. All three simulations use the same controls to switch views. You can change your viewpoint with the keyboard or the trim switch on the ThrustMaster Flight Control System or the CH FlightStick Pro.



The *Hornet* cockpit offers some features that don't appear in *Falcon 3.0*. It has three multifunction displays instead of one. To accommodate the extra instrumentation, the Hornet cockpit is divided into two parts.

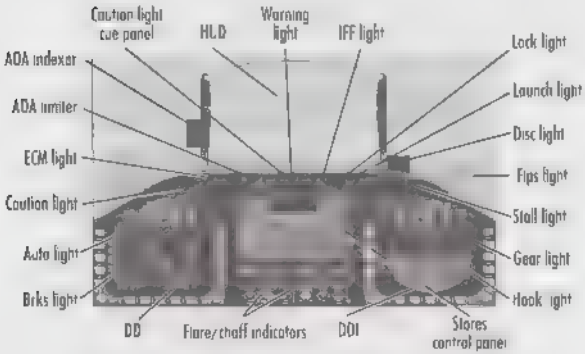
By default, the upper part of the cockpit appears on your screen. This view shows the head-up display and the two digital display indicators. It's called the look-up view, and it corresponds to the normal forward view in *Falcon 3.0* or *MiG-29*.

The head-down display called the HDD horizontal indicator and shows whether the HSI horizontal situation indicator or the COMED combined map electronic display)

To switch between the look-up and look-down views press the [0] key or the [5] key on the numeric keypad

Cockpit look-up view

Here's the standard cockpit view in *Hornet*



The head-up display HUD and the two digital display indicators DDIs are complex enough to warrant their own chapters See Chapter 3 Multifunction Display Group and Chapter 4 Head Up Display for more

Stores control panel

This panel shows which systems are active and which weapon is selected

Launch light

This light comes on when an enemy launches a radar missile at you.

Lock light

If your plane is locked in a radar beam, this light comes on.

Warning light

This light is a visual counterpart of the "WARNING WARNING" message that comes from the Voice Message System.

Caution light

This light is the visual equivalent of the "CAUTION CAUTION" message from the Voice Message System.

Flare/chaff indicators

These show how many flares and chaff pods you have left.

Stall light

This light comes on if your airplane has stalled.

Auto light

If you switch to autopilot, this light turns on.

Brks light

When you use your speed brakes, you see this light.

Disc light

During takeoff, this light indicates that the nosewheel steering system has disconnected. It comes on when your plane leaves the ground and stays on until you raise the landing gear.

Gear light

This light shows that the landing gear is down.

Hook light

This light shows that the tailhook has been lowered in preparation for landing on an aircraft carrier.

Caution light cue panel

This group of lights shows you which systems are damaged. Although the individual lights here don't have labels, their placement is the same as in the caution light panel in the right view described below. For instance, if you know where the ENG light is on the caution light panel, you can recognize it when it comes on in this view. The advantage is that you don't have to switch views.

Angle of attack indexer

This gauge helps you maintain the proper approach angle when you land the plane.

ECM light

This light comes on when you activate your electronic countermeasures pod.

Flps light

The flaps automatically configure themselves for takeoff, cruise and landing. The flaps move to landing configuration when you lower the landing gear. When the gear is raised, they move to normal configuration. The gear should be down and the flaps light should be off for takeoff configuration.

IFF light

The Identify Friend or Foe light helps you identify another aircraft. For details on the IFF system, see your *Falcon 3.0 Flight Manual* or your *MiG-29 Reference Manual*.

G-limiter light

This light shows the status of the Hornet's G limiter. If you need to perform a maneuver that would produce more than the 7.5 Gs that the limiter allows, you can

turn the G-limiter off. Press [G] the letter to override the G-limiter.

AOA limiter

The Hornet is capable of achieving angles of attack in excess of 90°. The NAVOPS manual specifically prohibits maneuvers with an AOA of more than 25° except in certain specific cases or emergencies. The maneuverability of the aircraft in this state is dangerously reduced.

When engaged, the AOA limiter limits the Hornet's angle of attack to 24°. When this system is active the AOA limiter light is red. If the light is green, which is the default condition, the system is disengaged. Turn the AOA limiter on or off by pressing [Shift] [O].

The AOA limiter does not exist in the real F/A-18. We have included it in this program because it is not yet possible in a computer simulation to provide the physical sensations that warn a real pilot when the aircraft is approaching dangerous angles of attack.

Cockpit look-down view

Here's the lower half of the *Hornet* cockpit.



You get this view when you press the [K] key or the [5] key on the numeric keypad. In the look down view you can see both digital displays, indicators and the horizontal indicator described in the next chapter. This view also contains a number of backup instruments that duplicate functions of the HUD and DDIs.

Standby radar warning receiver (RWR)

If you have one of the DDIs set to RWR mode, this display is blank. Otherwise it shows the same information that you would see on a DDI in RWR mode. It has lights to warn of radar lock (yellow) and missile launch (red).

Note to *Falcon 3.0* players. The RWR is the Navy equivalent to the Air Force's TWI (threat warning indicator). Keep this in mind when you see the RWR mentioned in this manual.

Standby airspeed indicator

This gauge shows the airspeed in hundreds of knots.

Standby fuel indicator

This information is also available on the DDI in engine monitor mode. The standby gauge shows the remaining fuel in thousands of pounds.

Standby attitude display indicator (ADI)

This "artificial horizon" shows the plane's pitch and roll attitude relative to the ground. It works like the ADI mode of the DDI.

Standby barometric altimeter

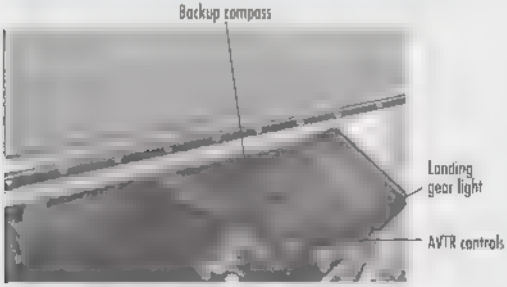
This gauge is a "three needle" altimeter. The short, medium and long pointers show altitude in multiples of 10,000 ft, 1,000 ft and 100 ft respectively.

Standby rate of climb indicator

This gauge shows your vertical speed in thousands of feet per second. The scale runs from -50,000 ft/min to +50,000 ft/min.

Left view

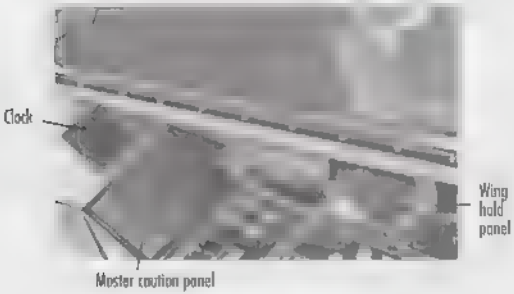
When you look out the left side of your Hornet, here's what you see:



The AVTR controls work just like the ones on Falcon 3.0 and MiG-29. Each instrument on the AVTR has a backup compass. The landing gear light glows red when the gear is up and green when it's down.

Right view

Here's the view from the right side of your Hornet:



The Wing Hold panel shows you the status of your F/A 18's wings. The Hold light is normally lit green. When you fold the wings (by pressing Ctrl W), the Fold light will illuminate red. In addition, the analog clock displays the time of day.

One or more of the 15 caution lights will illuminate if your aircraft sustains damage. Each of these lights indicate that the system in question is inoperable. These lights correspond in position to the caution light cue panel in the front view.

FLAP caution light

The FLAP caution light indicates that your wing flap mechanisms have been damaged and are frozen in their current state.

BURN caution light

When the BURN caution light illuminates, it means that your afterburner has been damaged. Your plane will immediately drop from any afterburner stage to 100% military power. You will have a more difficult time escaping from an area quickly or completing high-speed maneuvers.

RADAR caution light

The RADAR caution light turns on when the radar display is no longer operational. You will no longer be able to achieve a radar lock on air-to-air and air-to-ground targets. Heat seeking missiles, however, will still be able to lock on.

FUEL caution light

When the FUEL caution light turns on, it indicates that you have a leak in your fuel system, although not the severity of the leak. Check the Engine monitor to see how much fuel is being lost.

HUD caution light

The HUD caution light will illuminate when your HUD has been completely knocked out. You will no longer see any HUD information so you must rely on secondary instruments.

NWS caution light

The NWS caution light comes on when your nosewheel steering system has been damaged. Nosewheel steering damage is usually caused by not following required takeoff procedures, landing too hard, or being hit by enemy gunfire. When the NWS system has been damaged you will have a hard time steering the plane once it has landed.

STORE caution light

The STORE caution light indicates that you will be unable to launch or jettison any external stores you have loaded on your F/A-18. This includes all air-to-air and air-to-ground weapons, external fuel tanks and ECM pods. If you are planning on fighting, you have only the use of your cannon.

OX LO caution light

The OX LO caution light illuminates when enemy gun fire pierces your Hornet's canopy. This damage causes a drop in cabin pressure, which limits your maximum ceiling to 10,000 ft. Flying above this will cause you to blackout, even if you are flying straight and level.

ECM caution light

The ECM caution light comes on when your ECM pod has been destroyed. You will no longer be able to turn on your ECM pod to spoof enemy missiles and radar.

ENG caution light

The ENG caution light indicates that your engine has been damaged beyond repair. Engine shutdown causes your engine to drop to 0% RPM and your airspeed will begin to decrease.

JAM caution light

The JAM caution light means that your cannon is jammed and no longer works. When your gun is jammed only your internal gun is damaged. Any other weapons are not affected.

NAV caution light

The NAV caution light will light up if the map display is no longer functional. In addition you will no longer be able to see the waypoint carets on your HUD. Without the map you will have to rely on the autopilot and terrain landmarks to navigate.

BRAKE caution light

The BRAKE caution light indicates that your speed brakes have been damaged are now stuck in their current position. If your speed brakes were open your speed brakes will be frozen in that position. If your speed brakes were closed you will not be able to rely on the speed brakes to slow your plane down.

ARM caution light

The ARM caution light illuminates when your arming mechanism has been damaged. This means that you will not be able to arm any air-to-air or air-to-ground weapons. Any cluster bombs, iron bombs and rocket launchers, though will still operate properly. However you will need to switch to your cannon for a dogfight or strafing.

WING caution light

The WING caution light turns on when your wings have been structurally damaged, usually due to pulling excessive Gs.

CHAPTER
3

MULTIFUNCTION DISPLAY GROUP



The cockpit of the Hornet features three large screens which can be adjusted by the pilot to show a variety of information. These Multi-Function Displays (MFD) group comprises two digital displays and centers DDIs and the horizontal situation indicator (HSI). All three appear in the cockpit look-down view, while the look-up view shows the DDIs but not the HSI. See Chapter 2 'Cockpits' for information on switching views.



HSI

Of the three, the HSI is the most specialized. It's a navigational aid that helps you fly along the planned route between waypoints. The two LDDIs are more versatile. They show you information from a number of sources.

Digital display indicators

The two screens on the right and left sides of the Hornet cockpit are the digital display indicators. Both DDIs can display text and graphic data in five modes.

- Engine monitor
- Radar
- Maverick missile
- Radar warning receiver (RWR)
- Attitude display indicator (ADI)

You can switch either DDI to any one of these modes at any time. The only restriction is that you can't set both

DDIs to the same mode at once. For instance, you might want the right one to act as a radar display and the left one to show radar warning info for that unit. You couldn't have two engine monitors at the same time, however.

When you enter the cockpit, the left DDI shows engine data and the right one is the ADI. Together, the two displays give you the necessary information for takeoff.

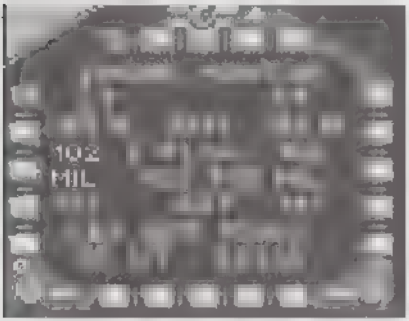
Each DDI has a keystroke control that switches to the next mode in the order they appear in the list above. Use these keys to switch DDI modes:

- Press Shift [to switch the left DDI to the next mode.
- Press Shift] to switch the right DDI to the next mode.
- Press Shift K to swap displays between the two DDIs.

The following sections describe the five DDI modes.

Engine monitor

This text display shows the state of the Hornet's twin engines.



On this display, lines three through six show readings for the individual engines. Each line contains one of the function codes in the list below. Numbers on either side of the code correspond to the current readings for the respective engine. For example, if you see a line like this:

490 EGT 500

it means that the engine gas temperature of the left engine is 490 while the right engine is operating at 500.

Here are the engine monitor codes.

RPM

Revolutions per minute. When the engines are running on military power, the RPM scale shows the number of turbine revolutions from 0 to 95% for military power. The number indicates actual RPM and not a percentage.

F FLOW

Fuel consumption flow in pounds per minute.

NOZ POS

The nozzle position of the engine. The possibilities are MIL for military power and AB1 through AB5 when the afterburners are lit (each AB number corresponds to an afterburner stage).

EGT

Engine gas temperature in degrees Celsius.

The two remaining lines show information that applies to the aircraft as a whole:

LBS REM

Pounds of fuel remaining.

GRS WT

The airplane's gross weight.

Radar

You can configure the radar in *Hornet* for the level of realism you want. Use the CONFIG display in the War Room to choose one of the three radar types.

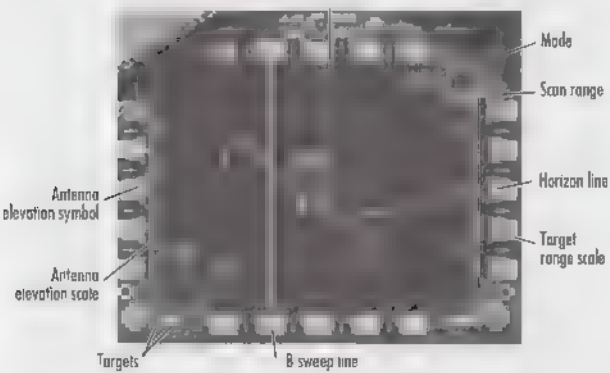
If you're a beginner, select one of the simplified radar types, the Situation Awareness Display (SAD) or the Situation Awareness Mode (SAM). If you want the radar in the game to behave as accurately as possible, choose the High Fidelity Radar (HFR) configuration.

In all three cases, the radar works like it does in *Falcon 3.0*. The display and the terminology are a bit different though. See the following sections for details.

Note: If you're used to flying the MiG-29, consult your *Hornet Briefing for MiG-29 Pilots* first.

Display

Here is a picture of the Hornet radar:



In this illustration, the radar is in Scan mode, one of the four modes of the High Fidelity Radar. Scan mode has one unique feature on the display, the **B-sweep**

line. This line moves back and forth as the radar antenna moves through its horizontal range.

The rest of the symbols on the above illustration appear in all radar modes in *Hornet*.

Mode

The current operating mode of the radar. The possibilities include:

- **SAD:** Situation Awareness Display
- **SAM:** Situation Awareness Mode
- **SCN:** Scan
- **TWS:** Track While Scan
- **STT:** Single Target Track
- **ACM:** Air Combat Mode

Scan range

The range of the radar in nautical miles. The four possible ranges are 10, 20, 40 and 80 nm.

Antenna elevation symbol

This mark shows the vertical position of the antenna.

Horizon line

This line moves as you maneuver the aircraft. Its vertical position indicates your vertical attitude. The horizon line position indicates your pitch attitude relative to the fixed velocity vector.

Target range scale

Use this scale to judge distances on the radar display. It is divided into four parts, each representing a quarter of the current scan range.

Antenna elevation scale

The antenna elevation symbol moves along this scale as the radar antenna scans up and down. The scale goes from +30° to -30° with 0 at the center. It is divided in 10° increments.

Target

Aircraft within radar range appear as inverted "V" symbols on the radar display. When you lock the radar onto a particular target, it flips over into a "U" shape.

Acquisition cursor

These two vertical lines let you select your target from the various targets on the radar display. Two numbers appear above the acquisition cursor. The top one is the minimum altitude coverage of the radar scan pattern. The bottom one is the maximum altitude coverage.

HFR modes

The High Frequency Radar has four submodes, each appropriate for a particular stage of an engagement. You switch between these modes as you fly and encounter enemies. HFR includes the following modes:

Scan (SCN)

SCN mode searches in front of your aircraft within the set distance.

Track While Scan (TWS)

TWS modes keeps scanning, but tracks a particular target and displays information about it.

Single Target Track (STT)

STT mode locks the radar onto one target, excluding all other targets.

Air Combat Mode (ACM)

ACM mode finds the first target within 10 miles and switches to STT mode.

Choosing a target

When you search an area for enemies, you typically begin with the radar in Scan mode. If your radar is not already in this mode, press F5.

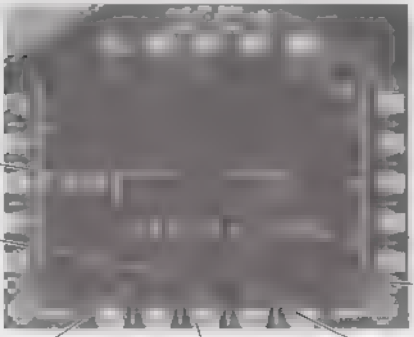
While you're scanning, you may want to take a closer look at a particular target. To set up a target for tracking, first move the acquisition cursor over it by pressing the T key. Every time you press T, the cursor moves to the next target on the radar.

When you have the target you want in the cursor, press Z to switch to TWS mode. In this mode, the radar continues to scan other targets, but it pays close attention to the one it's tracking. If the target moves out of range, the radar automatically increases its range to keep the target in sight. If the other targets leave the radar, it automatically switches to STT mode.

When you're close to a particular target, pressing a third time locks it up in STT mode. The second time you press Z to get to STT mode varies depending on the current state of the radar.

- If you're in Scan mode, move the acquisition cursor to a target and press the Z key twice.
- If you're in TWS mode, press Z once.

In some cases, the radar goes into STT mode automatically. As noted above, this happens if you're in TWS mode and all the other targets go out of radar range. Also, the Auto Combat Mode is basically a shortcut method for finding the closest target and switching to STT mode.

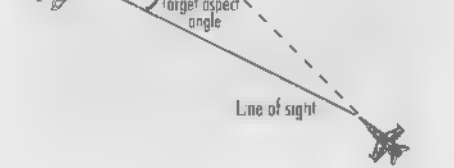


A number of changes occur in STT mode. They appear on the previous illustration.

The B sweep line will disappear because the radar is constantly interrogating the contact with no sweep. In STT mode the **tracked target** symbol changes from an inverted "V" to an upright "V" as this shows. The number on the left scale now displays the target's relative altitude in thousands of feet.

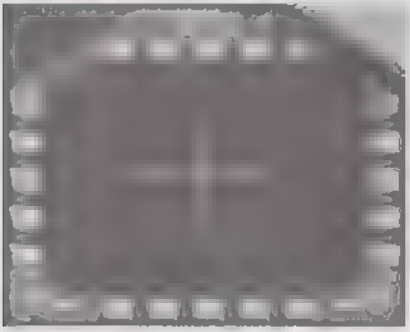
Two numbers appear on either side of the tracked target. The number on the left is the target's relative altitude to the nearest thousand feet above mean sea level. The one on the right is the target's airspeed in tens of knots.

A solid straight line or "stalk" appears on the locked target to show its **direction of travel** relative to you. The "stalk" is also known as the target aspect angle pointer graphic. It only displays the target's aspect angle. The target's aspect angle is the angle between your line of sight to the target and the target's velocity vector. It is used to tell if the target is coming at you or flying away from you. This is important since the Navy rules of engagement (ROE) specify different ROE for Air Force. This means, for example, that if the target's stalk is pointing straight down, the target is flying straight ahead of the target's tail, pointing straight up, the target is heading directly away from you regardless of where it is on the radar screen.



Maverick

To get this display on a DDI, you must have a Maverick missile selected as the current weapon. If you have another weapon selected, the monitor will be blank. Switch to the Maverick monitor by pressing Shift _ or Shift .



The Maverick DDI mode looks like the corresponding display in the F-16. An advantage Hornet players have over Falcon 3.0's is that you can view both the radar and the Maverick display simultaneously (one per each DDI).

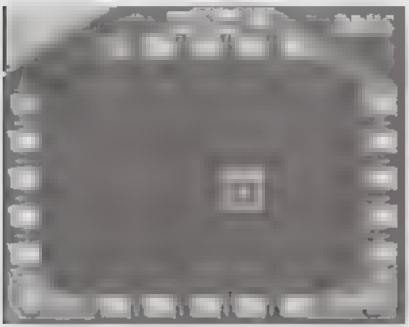
Note: If you're used to flying the MiG-29, consult your *Hornet Briefing for MiG-29 Pilots* first.

Radar warning receiver

The RWR display is a passive warning device that shows you the position of enemy aircraft radars, SAM radars and radar-guided missiles within approximately 30 miles. It works exactly like the F-16's threat warning indicator (TWI) in *Falcon 3.0*.

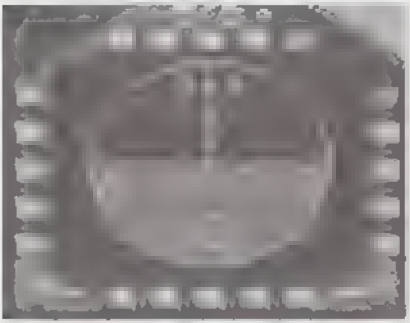
In addition to the DDI threat mode, the *Hornet* cockpit has backup warning systems. The cockpit's look-up view highlights both the gate enemy radar lock and missile launch. Also, there is a stand-by RWR display in the look-down view.

Note: If you're used to flying the MiG-29, consult your *Hornet Briefing for MiG-29 Pilots* first.



Attitude display indicator

This display shows the orientation of the plane relative to the horizon.



The ADI is divided into two parts representing the sky and the ground. The stationary line through the middle represents the wings of your plane. As you turn to the left or right the ADI's "artificial horizon" tilts accordingly. It rises or falls when you dive or climb.

The ADI is useful when the skies are overcast or whenever you can't see the real horizon.

Like the RWR, the ADI has a backup system. It's in the look down view to the right of the cockpit.

Horizontal indicator

The HI, is the third main function display in the cockpit. It's a navigation aid that helps you follow your flight plan. The HI uses information from a network of radio beacons to pinpoint the location of your aircraft. It converts this data into symbols and graphics that, in effect, let you look down on your plane from above.

The HI display operates in three modes.

HSI symbolic display

This mode places your Hornet at the center of a compass circle. When you turn, the compass rotates to show your current heading. The display also includes symbols for your waypoint bearing and other navigational parameters. These are explained below.

Map mode

In this mode the HI shows a color map of the terrain beneath your aircraft.

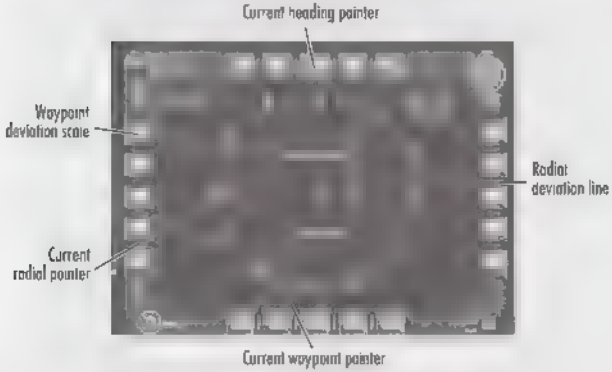
Combined HSI and navigational map mode

This mode superimposes the HSI symbols on top of the map.

To switch between the three modes, press C.

HSI (horizontal situation indicator)

In its default mode, the HSI looks like this



Your Hornet is represented by the aircraft symbol at the center of the display. The other main items of interest are

Current heading pointer

Also known as the "lubber line," this triangle remains fixed at the top of the display while the compass circle rotates to show where you're going.

Current waypoint pointer

This is the bearing to the current waypoint. If this marker is lined up with the current heading pointer, you know you're flying towards the waypoint. However, you may not be following your flight plan. The difference can be fatal.

Current radial pointer

This is the approach path, or "radial," to the waypoint. Although you can approach a waypoint from any direction, some are safer than others. Your flight plan specifies a particular radial for each waypoint.

Radial deviation line

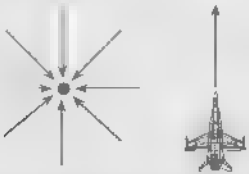
This arrow slides back and forth across the waypoint deviation scale. It shows the difference in degrees between the current waypoint bearing and the current radial heading.

Waypoint deviation scale

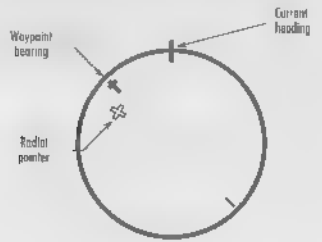
This scale extends from −10° to +10°. When you're following the radial to the current waypoint, the line is in the center of the scale. In other words, there are 0° of separation between the radial heading and the waypoint bearing.

Navigating with the HSI

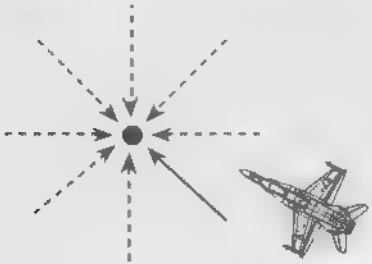
You can approach a waypoint from any direction. In theory, every waypoint has an infinite number of approach paths or "radials." Here they are, along with a Hornet that's off course:



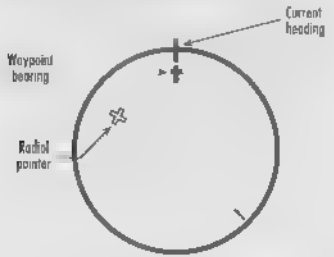
The arrows are the radials that lead to the waypoint. If the off course pilot at this diagram glanced at the HSI, it would look something like this.



If the pilot wants to head towards the waypoint by the straightest route, it doesn't matter which radial he or she picks. All the pilot has to do is turn in the right direction.



In this situation the pilot doesn't need the HSI. The HUD's waypoint indicator gives the necessary information to make the turn. However, for purposes of illustration, here's what the HSI would look like after the pilot turned toward the waypoint:



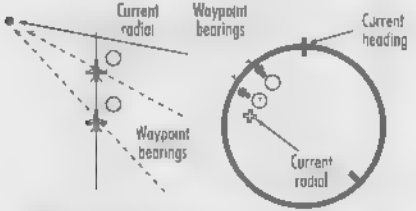
Notice that the waypoint bearing marker is lined up with the "rabbit ear" or current heading symbol. Also notice that the radial marker isn't lined up.

Your flight plan specifies a particular approach path for each waypoint in a mission. There's a good reason for choosing one radial over the others. The shortest path to the waypoint may also be the path that takes you over a SAM emplacement to certain doom.

This is where the HSI comes in handy. It helps you get to the waypoint by following the radials in your flight plan. In general, when you navigate using the HSI, the goal is to make sure your plane's heading, the three pointers, line up. When they're aligned, you know that you are flying towards the waypoint along the path created by your flight plan.

If you're off track, try this procedure:

1. Maneuver so that the pointers to the current heading and waypoint bearing are on the same side of the radial pointer.
2. Turn and fly towards the radial. The next diagram shows the pilot's actions and the results on the HSI.



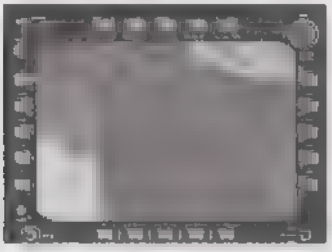
Now see what happens as the pilot flies from position one to position two. As the aircraft's waypoint bearings converges on the radial, the waypoint marker on the HSI moves towards the radial pointer. When the radial reaches one, as you can see, you know you're headed in the right direction.

3 When the radial and waypoint markers are aligned, turn to bring them both in line with the heading pointer. Here's the outcome:



Map mode

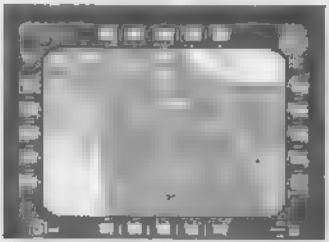
When you switch the HI to map mode it shows a color map of the terrain beneath your aircraft.



The HI map mode is identical to the on-board map display in *Falcon 3.0*. The red plus sign is your plane's position, the green triangle is your current waypoint, and the red "X" is your home base.

Combined HSI and navigational map mode

Finally, here's the HI mode that combines the other two displays. When you are in this mode, you can see where you are relative to the world using the map and where you are relative to your flight plan using the HSI. This combined mode, along with the ADI, lets you fly heads-down.



CHAPTER
4

HEAD-UP DISPLAY



The head-up display (HUD) shows vital information from your flight and weapons control systems. Although this data is available on other instruments, the HUD puts it where you really need it: right in front of you. You can read the data on the HUD while looking through the canopy of your Hornet. In a dogfight, the HUD can save your life.

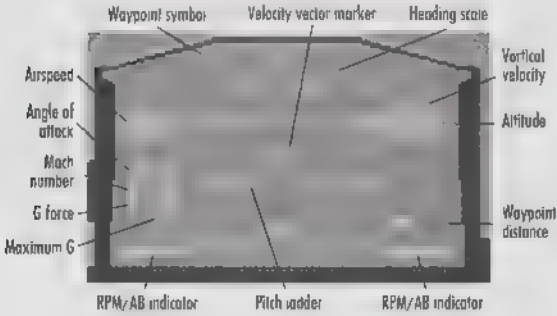
The information on the HUD changes as you go from job to job in a mission. The list below shows the various HUD modes:

- **Air-to-air mode**: Press Enter to select an air-to-air weapon and activate this HUD mode.
- **Air-to-ground mode**: Press Backspace to activate a bomb or other air-to-ground weapon and see this HUD mode.
- **Instrument Landing System (ILS)**: Press ▢ to activate ILS landing mode.

The following sections describe these modes, as well as the basic HUD features that stay the same from mode to mode.

HUD basics

The following illustration shows the features that are displayed in all HUD modes.



These features include:

Heading scale

The aircraft's magnetic heading appears on a 360° scale that rotates as you turn the aircraft. A small, stationary cross above the scale marks the current heading.

Waypoint symbol

This symbol indicates the direction of the current waypoint. When the waypoint is centered on the heading scale, your plane is flying directly towards the waypoint.

Airspeed

The number in this box is the airspeed as calibrated by the air data computer (ADC).

Altitude

This box shows barometric altitude, your altitude above sea level, or radar altitude, your altitude above the ground. You can toggle between barometric and radar altitude by pressing Ctrl+A. The switch sets the radar preset in alt. R appears next to the altitude display to indicate that it contains radar data. If the radar altimeter is out of range, or data is not available, flashing lines replace your HUD's altimeter display. Barometric altitude when you are above 50,000 ft. Above 59,999 ft, the display changes to XXXXX.

Angle of attack

The alpha symbol, α, on the HUD indicates the angle of attack in degrees.

Mach number

The letter M indicates the speed of your aircraft as a fraction of the speed of sound.

G force

The letter G indicates the current strain of G forces on the aircraft. The readout will never go above 9.9 Gs, even if you override the G limiter and pull more Gs.

Maximum G

This indicator measures the highest Gs your plane has pulled during the flight. It is activated the first time you exceed 4 Gs.

Velocity vector marker

This indicator marks the spot in the outside world that your plane is flying towards. When you make directional changes to the aircraft, the velocity vector marker will move to show you the direction you are currently heading.

Vertical velocity

The number above the altitude box is the rate in tens of feet per minute at which your plane is climbing or descending. A negative number indicates descent.

Pitch ladder

You can see the vertical flight path of your plane by noting the position of the vertical velocity indicator on the pitch ladder. The "rungs" on the ladder mark your position relative to the horizon in 5° increments. The pitch ladder rungs "bend" in relation to the vertical velocity.

Waypoint distance

This number is the distance to the next waypoint. The code next to the number shows your current goal. For example, if it says "D4," you're heading towards the third waypoint.

RPM/AB indicators

The values on the lower left and right corners show the status of the respective engines. A number from 0 to 100 indicates engine RPM as a percentage of maximum (MIL). The number on the left shows the current RPM status. The number on the right shows the throttle setting. The number on the left should catch up to the number on the right as the engine spools up. If you

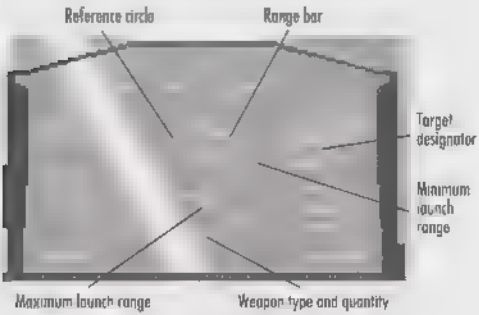
engage the afterburners, the display changes to show the burner stage. Each engine can have a value from AB1 to AB5.

Air-to-air HUD mode

In air-to-air (A-A) combat, the HUD helps you find your target and bring it into the range of your weapons. There are two variations on this HUD mode, one for the Sidewinder and AMRAAM missiles and one for the cannon. When you select one of these as your current weapon, the HUD automatically switches to the appropriate display mode. It shows targeting information for the missile or gun and tells you when conditions are right to fire your weapons.

HUD for A-A missiles

The next illustration shows the A-A HUD as it appears when you select the Sidewinder missile as the current weapon. The HUD display for the AMRAAM missile is basically the same.



This version of the HUD contains the following elements:

Target designator (TD)

This box identifies the radar's line of sight to the target. The TD appears in all three A-A weapon modes. When the target moves out of the HUD, the TD disappears. Instead, the HUD displays the target locator line and the diamond X symbol (see below).

Weapon type and quantity

The code on the left indicates which A-A weapon is active. The codes are:

- **9P:** AIM-9P "Sidewinder" rear-aspect infrared missile
- **9M:** AIM-9M "Sidewinder" all-aspect infrared missile
- **120:** AIM-120 AMRAAM missile
- **GUN:** M61A1 20mm cannon

The digits beside the code indicate the number of remaining missiles or cannon rounds.

Reference circle

This circle, centered around the airplane's waterline, acts as an aiming point for shooting missiles.

Range bar

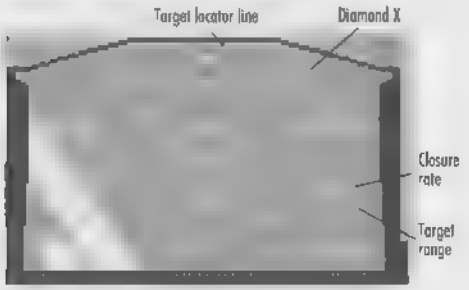
This mark moves around the inside of the reference circle. Its position indicates your distance to the target relative to the minimum and maximum ranges for the selected weapon. Your goal is to maneuver the plane so that the range bar is between the two triangles outside the reference circle.

Minimum launch range

This triangle at the 2:30 position outside the reference circle marks the lower limit of the range for successfully launching your missile. If the target is any closer, the missile can't hit it.

Maximum launch range

This triangle, at the 6:00 position, marks the upper limit of the range for a successful missile launch. If the target is any farther away, the missile is unlikely to hit it.



Target locator line

When the radar is tracking a target outside the HUD, this line points in the target's direction.

Diamond X

As the target passes out of the HUD, the target designator is replaced by the diamond X symbol. This symbol stays at the edge of the HUD as a steering aid. When the target re-enters the HUD, it appears at the position of the diamond X.

Target range

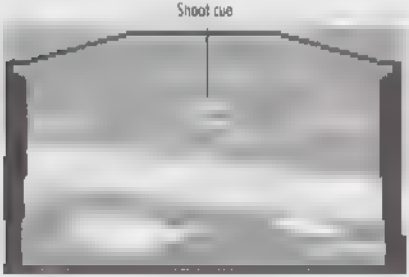
This is the distance between you and the target to the nearest tenth of a nautical mile.

Closure rate

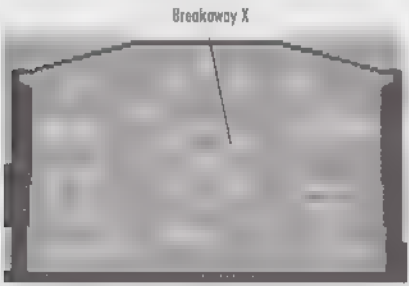
This number is the speed in knots at which you and the target are approaching or moving away from each other. A positive number is the closing velocity, or the rate at which the two aircraft are getting closer. A nega-

tive number is the "opening velocity" or the rate at which the two aircraft are getting further

In addition to the above items, the HUD provides two more cues to let you know if the conditions are right for firing your weapon. Here's one:



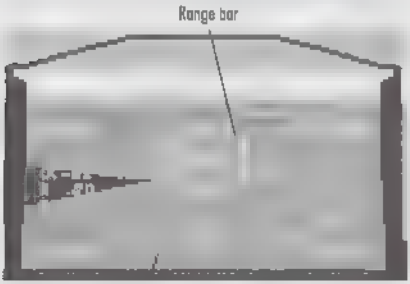
The **shoot cue** comes on when the target is in range and every other launch conditions satisfied. When you see it, pull the trigger



The "X" means you're inside the selected missile's minimum range. It tells you to break off the attack, choose a shorter range weapon, or maneuver your plane until the enemy is vulnerable

HUD for A-A cannon

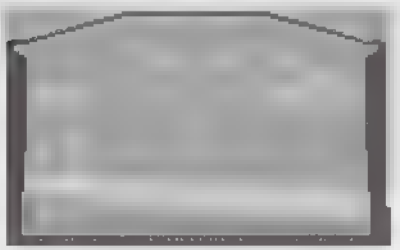
If you pick the cannon as your weapon of choice for airborne combat the HUD display looks like this.



The number under the GUN code tells how much ammunition you have. When you run out this number reads "XXX."

Inside the reference circle a range bar shows the distance to the target. The bar counts down from 2,000 ft counterclockwise where each clock unit is 200 ft. For example, if the range bar is 3 o'clock, the target is 3,000 ft away. The effective range for the gun is 3,000 ft although the maximum range is 5,000 ft.

If you turn off the radar or it is disabled you can still aim and fire the cannon. Naturally the HUD won't display a range bar or other targeting information that comes from the radar. Instead it shows a simple aiming reticle that looks like this:



Air-to-ground HUD mode

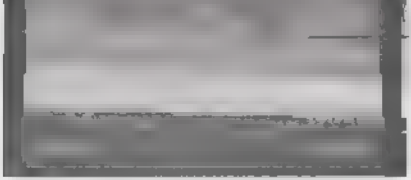
Since the Hornet is primarily a strike aircraft, you spend a lot of time dropping bombs and launching air-to-ground (A-G) missiles. When you select an air-to-ground weapon, the HUD switches to one of its four A-G modes:

- **Maverick (MVRK)** For launching Maverick missiles and guided bombs.
- **Computer controlled release point (CCRP)** For dropping dumb bombs or guided bombs.
- **Strafe (STRF)** For firing the cannon or launching unguided rockets.
- **Anti-radiation (A-R)** For launching anti-radiation guided missiles, such as the Shrike or HARM.

For the most part, these variations on the HUD behave like their counterparts in *Falcon 3.0*. The displays for the Maverick and anti-radiation missiles show cross hairs that lock on to the target and follow it across the HUD.

Note: If you're used to flying the MiG-29, consult your *Hornet Briefing* or *MiG-29 Pilot's Notes*.

When you choose the cannon or rocket pod, the HUD displays a reference circle for aiming purposes.



If your radar is active the **range bar** indicates the slant range or distance to the ground inside the reference circle. The range bar counts down from 12,000 ft counterclockwise where each clock point is equivalent to 1,000 ft. Once the bar moves past the three o'clock position, you have a good chance of hitting the target. The break X cue comes on when you're getting close to the ground. If you continue diving, you will hear the "PULLUP-PULLUP" warning.

When you choose one of the unguided bombs as the current weapon the HUD changes to CCRP mode.



In this mode the HUD contains the features that appear on the following list. Some appear in other A-G HUD modes.

Mode

Aside from CCRP, the modes can be MVRK (Maverick), STRF (A-G cannon or rockets) or ARM (anti-radiation missile, such as the Shrike or HARM).

Weapon type and quantity

The code at the top indicates the current weapon. The bottom number is the remaining bombs, missiles or cannon rounds.

- 65b Maverick AGM 65B TV guided missile
- GBU GBU 15 guided bomb unit
- 65D Maverick AGM 65D infrared guided missile
- DCR BLU-107 B Durandal anti-runway bomb
- 82H Mk82 high-drag iron bomb
- 82 Mk82 iron bomb
- 83 Mk83 iron bomb
- 84 Mk84 iron bomb
- CBU CBU 84 cluster bomb unit
- GUN M61A1 20mm cannon
- LAU LAU rocket
- 88A AGM 88A HARM anti radiation missile
- 45A AGM 45A Shrike anti radiation missile

True airspeed

When the HUD is in CCRP or strafe mode the true airspeed appears preceded by the letter "T"

Computer controlled release point (CCRP)

The CCRP marker shows the point where a bomb will hit if you designate the target at that point. If the impact point is outside the HUD no CCRP marker appears

Displayed impact line

Before you designate a target, this line appears as a steering aid. Use it to manoeuvre the aircraft so you can position the CCRP cross over the target

Once you designate a target placing the CCRP marker on it and by pressing the Spacebar or joystick button, the HUD changes to the following display

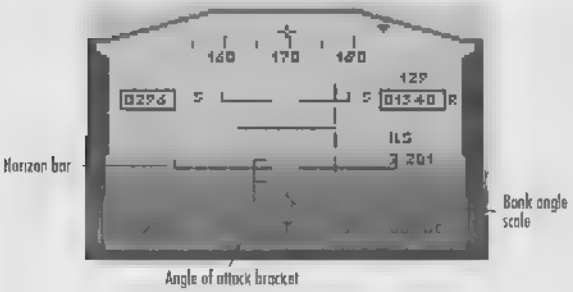


The **azimuth steering line** (ASL) is always parallel to the horizon. It's a steering aid that helps you keep the aircraft level as you approach the weapon's release point. The **displayed impact line** (DIL) helps you stay on course to the weapons release point.

As you approach the target, the **release cue** moves down the DIL. When it intersects the ASL, the computer will automatically release the bombs.

Instrument landing system

The instrument landing system helps you land the Hornet. It comes on when you press [ ]. Here it is.



The unique features of the ILS are:

Horizon bar

This "artificial horizon" helps you orient the plane during landing.

Angle of attack bracket

The position of the velocity vector marker on this bracket indicates your approach angle. The center of the bracket represents the optimum approach angle. When the velocity marker is above the center line, your angle of attack is too high. When it is beneath the center line, your angle is too low.

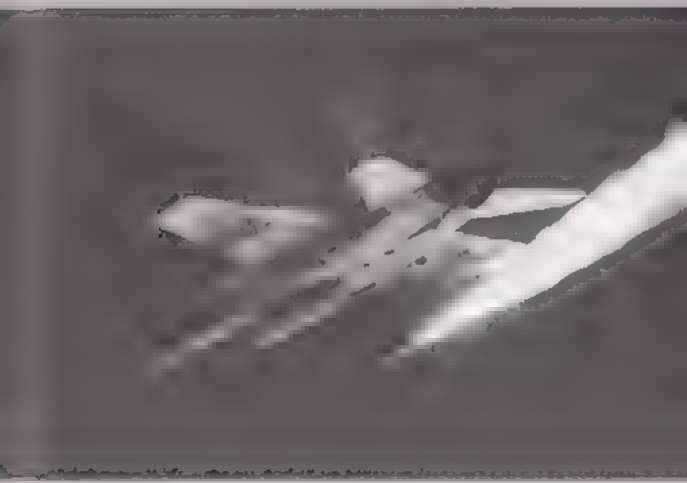
Bank angle scale

A triangular pointer moves across this scale to show your bank angle. The tick marks along the scale divide it at 5°, 15°, 30° and 45°.

For information on landing the Hornet, refer to the next chapter.

CHAPTER
5

AIR COMBAT TRAINING



As a veteran of *Falcon 4.0* or *MiG 29*, you already know how to fly a high-performance aircraft on your computer. You know the controls, the instruments and the tactics. Much of that knowledge carries over to *Hornet*. That means we don't have to tell you which key to press if you want to light your afterburners. You don't need us to point out the blinking enemy radar on the tactical display. That's the fun part of an offensive one.

However, you do need a chance to get familiar with your new airplane. This chapter provides the opportunity. It has four lessons that teach you the basics of air combat in your Hornet:

- Lesson 1: Takeoff and landing
- Lesson 2: Air-to-air missile practice
- Lesson 3: Air-to-ground combat: start weapons
- Lesson 4: Air-to-ground combat: bombing

Each lesson is a training mission that takes place in the Red Flag area. To make sure these missions work correctly, you need to set the Difficulty level of the game to Intermediate. Click on the CONFIG button in the War Room and, when you get to the Difficulty Levels screen, click the Intermediate button. Then click Done to get back to the War Room.

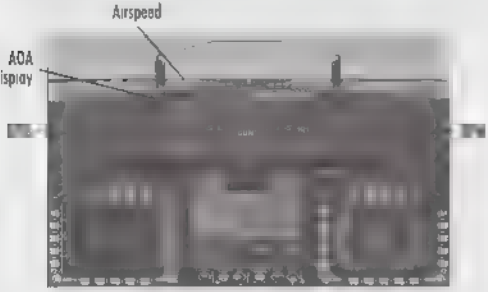
To fly one of these missions, first choose a squadron in the War Room screen, then go to the Red Flag area. If you need more information on Red Flag, refer to your *Falcon 3.0* or *MiG 29* documentation.

The instructions for each lesson direct you to load a particular mission file. You may recognize these missions if you worked through the Air Combat School curriculum in the *Falcon 3.0* or *MiG 29* documentation. This chapter takes you through some of the same procedures, but it also points out unique points about your F/A-18 Hornet.

Lesson 1: Takeoff and landing

In this lesson, you practice the two most crucial skills that any pilot must master. You begin on the runway of Nellis Air Force Base. The lesson shows you how to take off, circle the airfield and land on a runway.

Once you're in Red Flag, load the file REVIEW1 and go through the standard procedures for assigning pilots and armament. When you click the Takeoff button on the Armament screen, here's what you see:



Notice that one of the DDIs is in Engine monitor mode. Since this mode is particularly useful during takeoff and landing, it appears by default.

Take off

Open up the throttle until the RPM indicators on the bottom of the HUD read "100 100." Then light the afterburners. When the airspeed indicator on the HUD reaches 150 knots, pull back on the stick. Raise your landing gear and climb to 15,000 ft. Then level off and cut the afterburners.

Circle the airfield

You're flying south away from Nellis. The heading indicator on your HUD should read 180°.

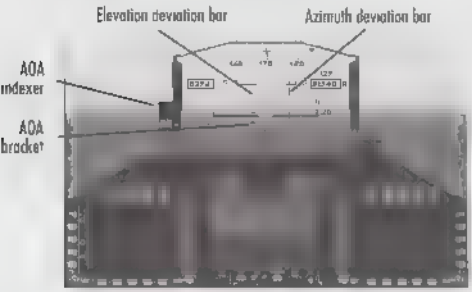
On this short mission, you'll turn your F/A-18 to the left and circle around until you're approaching the landing strip. Follow these steps:

1. Bank left until your heading reads 90°. Then level off. Fly in this direction for a minute or so.
2. Bank left again until your heading is 0°. As you fly in this direction, look out the left side of your cockpit and watch the airfield go by. Continue flying straight on for a minute or so.
3. Bank left again until your heading is 270°. Again, look out the left side of your cockpit to keep an eye on the airfield.
4. When the airfield is almost at the center of your field of view, bank left to a heading of 180°.

Landing

This lesson uses the standard "racetrack method" for landing the plane. You'll line your F/A-18 up with the runway, fly over it, circle around by making two 180° turns and come in for your final touchdown. At this point, you should be headed south, towards the airfield. To make your approach and land the plane, follow these steps:

1. Switch the HUD to ILS mode. The cockpit should look like this:



2. Descend until you are 10,000 ft and 10 miles from the runway. You should be traveling at about 450 kts. As you begin your approach, use the ILS as a guide to correct the horizontal and vertical orientation of your plane. When the elevation deviation and azimuth deviation bars form a cross, you're on target.

3. Continue your approach. When you're five miles from the runway, slow to 270 kts. At two miles, slow to 230 and drop your landing gear. The flaps will come on automatically.

4. When you're aligned with the runway, fly over it and make a 180° turn. Fly back parallel with the runway, maintaining an 800 ft altitude and passing it by about two miles. Then, make another 180° turn and head back in.

5. Again, maneuver the plane so that the elevation deviation and azimuth deviation bars form a cross. Also, keep an eye on the two angle of attack indicators. The AOA indexer, the three lights to the left of the HUD, works like the one in Figure 3-1. The AOA bracket on the HUD performs a similar function. Look at the position of the velocity vector marker next to the bracket. When the marker is next to the bracket's center line, you have the correct angle of attack. If you want to see the AOA in degrees, switch the HUD out of ILS mode by pressing Enter or (Backspace). The AOA will appear on the left of the HUD.

6. With the two bars on the ILS forming a cross, reduce your airspeed until it's between 160 and 190 kts. Make sure your angle of attack is at least 8°.

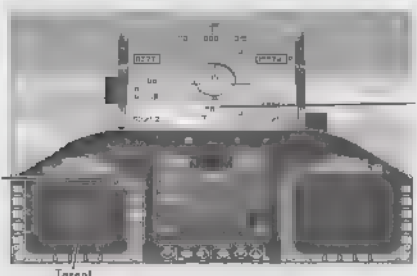
7. Aim the velocity vector for the beginning of the runway and come in for a landing. When you touch down, lower your nose and reduce speed. Engage your speed brakes. When you come to a stop, switch on your wheel brakes.

Lesson 2: Air-to-air combat

In this lesson, you confront a Mirage III fighter. As you chalk up a kill, you'll learn about the F/A-18's air-to-air missiles and HUD display. You'll also get some experience with the radar display on the DDI. Although you can fly this mission using one of the simplified radar modes, it's a good idea to practice with the High Fidelity Radar. The High Fidelity Radar contains a lot of information that you need to absorb at a glance during combat. To use the High Fidelity Radar, go to the Difficulty Levels screen and check the RADAR switch until you see the HFR setting.

When you're ready to shoot down an enemy fighter, go to the Red Flag area and load the file LESSON5. After you click Takeoff on the Armament screen, follow these steps.

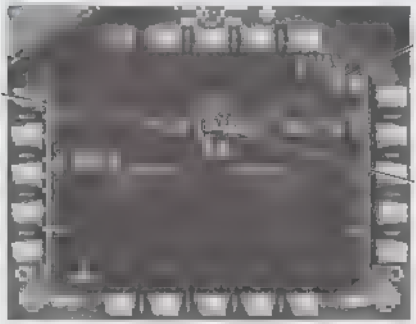
1. Press Enter twice to select the Sidewinder AIM-9M as your weapon.
2. Press F5 to set your radar to Scan mode. At this point, your screen should look something like this:



The upside-down "U" symbol is the Mirage.

3. Press T to move the acquisition cursor over the target symbol.

4. Press Z to switch the radar into Track While Scan mode. Here is the result:



Notice that the target changes to an upright "U" symbol. Also, keep an eye on the numbers that appear around the target. The one on the right is the target's airspeed in knots, rounded to ten. The left-hand number is the target's altitude to the nearest thousand feet above sea level. The numbers above and below the acquisition cursor represent the radar's minimum and maximum altitude coverage, respectively.

5. Follow the Mirage until it comes into firing range. As you maneuver, watch how the range symbols on the HUD change. As you approach the Mirage, the range bar which is the reference circle moves counter-clockwise. When it moves between the two triangles outside the circle, you're in range. Maneuver the TD box inside the reference circle for the optimum missile PK (probability of kill).

6. When the word "Shoot" appears on the HUD, fire your missile.

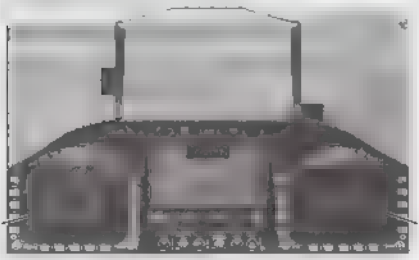
Once you've destroyed the Mirage with the Sidewinder missile, try working through the mission again using the AMRAAM.

Lesson 3: Air-to-ground combat (smart weapons)

Here's your chance to practice the kind of mission the Hornet was designed for. In this lesson, you'll use the Maverick missile to eliminate some tanks on the practice range at Nellis Air Force Base.

As usual, start off by going to the Red Flag area. Choose the file LESSON7A and proceed to takeoff. Then follow these steps:

1. Press Backspace to select the Maverick missile as your current weapon. Switch your right DDI to the Maverick monitor so it displays the characteristic cross hairs.
2. Press F7 to switch the other DDI to Ground Map (GM) radar. You can use the GM radar to locate ground targets as you do when you fly the F-16. The advantage of the Hornet is that you can see both the GM radar and the Maverick display simultaneously.



3. To see the terrain map of the territory beneath you, press 5 on the numeric keypad and then press C. The horizontal indicator switches to its color map mode. Press 5 again to return to the look-up view.
4. Point your nose down until you see a line of dots on the ground beneath you. These are the tanks you want to destroy. When you see them, maneuver the

airplane until the Maverick cross hairs on the HUD lock onto a tank.

5. Since the Maverick is a fire and forget missile you can immediately fire by pressing Spacebar and then target the next tank by pressing T. Fire on as many tanks as you can.

There are plenty of other targets on the Nellis range waiting for your Mavericks and other air to ground weapons. Try using your cannon and rocket pod too.

Lesson 4: Air-to-ground combat (bombing)

This is your opportunity to take out a major target a nuclear facility. You'll fly your Hornet along a predefined flight plan using your horizon and radar to help you navigate. When you reach the nuke plant, you'll reduce it to rubble with Mk84 iron bombs.

By the way, don't assume you'll be alone up there just because you're on a strike mission. You may get some extra practice in air to air combat on your way to or from the nuclear plant. Be prepared.

In the Red Flag area, select the file RTVIFW2. In this mission you begin on the runway at Nellis. Follow the directions below to complete your mission.

Take off and climb to 10,000 ft. If you need a review of takeoff procedures, refer to Lesson 1 earlier in this chapter.

2. Select the AIM-9M as the current weapon. If you encounter any hostile aircraft during this mission, engage and destroy them.

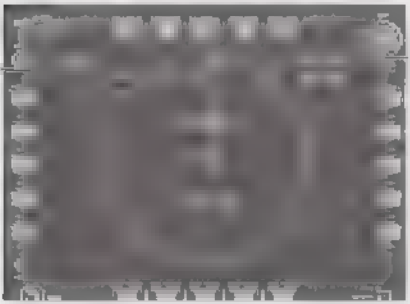
At this point you need to turn to head towards your first waypoint. Although you could use the waypoint marker on the HUD, try navigating with the horizontal situation indicator (HSI) for practice. The next few steps show how.

3. Press 5 on the numeric keypad to get the look down view, which includes the HSI. Here it is, showing your current heading of 180°.



Notice that the waypoint and radial markers are merged into a solid cross. On this session you don't have to follow a planned radial to your destination. Just point the plane towards the waypoint and fly. For detailed instructions on using the HSI to navigate using radials, see Chapter 3, Multifunction Display Group.

4. Turn to your new heading of 300°. As you bank the plane, notice that the HSI compass turns. Turn until the waypoint cross lines up with the heading marker at the top of the compass. When you're on track, the HSI should look something like this.

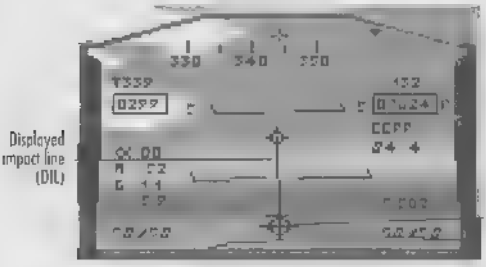


5. Press 5 again to return to look up view. Continue flying on this course until you reach the first way point.

6. At the way point change your heading to 0°. This course takes you towards the nuclear plant.

7. When you are within 10 miles of the plant, select the Mk84 bomb as your current weapon. The HUD changes to CCRP mode.

   Notice the displayed impact line (DIL). The DIL runs from the velocity vector to the center of the CCRP cross. Use the DIL for reference as you maneuver the plane to get the CCRP on target.



8. Head towards the nuclear plant, at an altitude of around 2,000 ft.

9. Maneuver the plane so that the CCRP is at the base of one of the towers. When it's in position, press Spacebar to designate the target.

10. Fly steady and watch the release cue move down the DIL.

11. When the release cue intersects the azimuth steering line, the bombs will drop.

12. Pull up and look out the back of your cockpit to see if you hit the target. If not, circle back and try again.

13. When you've destroyed the nuclear plant or run out of bombs, change course and head for home.

CHAPTER
6

CARRIER OPERATIONS



As the premier strike fighter of the U.S. Navy and Marine Corps, the Hornet is a carrier based aircraft. Almost all missions in *Hornet* begin and end on the USS Theodore Roosevelt.



Aside from the campaign missions in *Hornet*, the aircraft carrier also appears in Instant Action, In Red Flag, there is a carrier deck complete with arrestor wires and other landing aids. If you want to practice carrier based takeoffs and landings, these areas provide a good opportunity. Once you enter Instant Action, you can find the carrier 12 nm to your left on the Persian Gulf just west of Farsaka Island and southwest of Kuwait City. To try Field Carrier Landing Practice (FCLP) in Red Flag, select LESSONS from the available missions and use the simulated carrier deck at the airfield to land.

Carrier takeoffs and landings

Carrier operations are a major part of the Hornet's mission and capabilities. It is, therefore, essential that every Hornet pilot learn how to take off from and land on an aircraft carrier.

As you will see, takeoffs are not particularly difficult comparatively speaking. Landings, however, are quite tricky. Here, step by step, is everything you need to know to launch from and land on a carrier.

Takeoff

The takeoff procedure for carrier operations is fairly straightforward. As long as everything is done in the right order, anyone can do it even the first time through.

Preparation

To set up for a carrier launch, select a squadron and theater in the War Room. Since most Hornet missions are carrier-based in every theater except Pakistan, you can select any of the other theaters to set up a carrier launch.

Select the Commit button. The usual campaign screens will go by. Just skip on ahead through all the mission preparation screens. After all, this is just a practice launch, not a full-blown campaign mission. You may want to select the Clear All Weapons button on the Armament Screen to avoid any extra weight or drag.

Launch

To get enough lift to take off, the Hornet has to be traveling at around 150 kts. The deck of an aircraft carrier, however, is not quite long enough to allow the F/A-18 to reach takeoff speed on engine power alone.

Built into the deck of every carrier is an aircraft launching system called a catapult. With the aid of a catapult, and with the engines going at full afterburner, the Hornet can generate sufficient lift for takeoff.

Increase the throttle (press [+] and go to full afterburner (press [ ] to light the afterburner, then press [>] four times to get to full afterburner. Hold the stick all the way back and press [Alt] [C] to activate the catapult. You will accelerate rapidly. As soon as you reach the end of the deck, ease the stick back to neutral to maintain a climb angle of about 10°.

Now that you're free of the deck, retract the gear (press [G]). Be aware that your cockpit view will suddenly shift upwards when the gear retracts. When the gear is down, you are assumed to be close to the ground, so

the view shifts downward. Be careful not to pull the nose too high.

Congratulations, you're airborne!

Landing

Carrier landings are quite a bit more difficult than launches. The normal difficulties of landing are compounded by the fact that your airstrip is extremely short and is moving.

Preparing to land

In a campaign mission, your final waypoint defaults to "LAND". This final waypoint is usually the stern of the carrier. Use your horizontal situation indicator (HSI) to approach along the correct path.

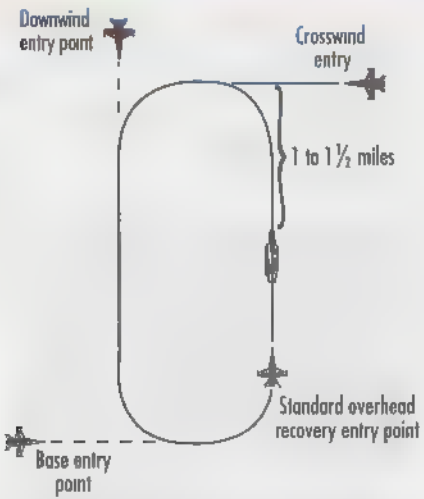
At 30,000 ft AGL, 20 nm from the carrier, begin descending at about 33 ft/sec; check your vertical velocity indicator on the HUD at a speed of Mach 0.8. At 8 nm you should be at 1,200 ft. Reduce your airspeed to 250 kts.

There are two ways to land on a carrier in *Hornet*. The first method is called a visual landing. In a visual approach, you enter a "racetrack" landing pattern around the carrier.

Visual (or pattern) landings

To properly execute a pattern landing, you must be able to see the carrier to orient yourself. If you cannot see the carrier, engage your ILS and follow it to the carrier for an instrument landing. See *Instrument or straight in landings* later in this chapter for details.

As you approach the carrier, the Carrier Air Traffic Control Center (CATCC) will direct you to enter the pattern at one of the points shown by the following diagram.



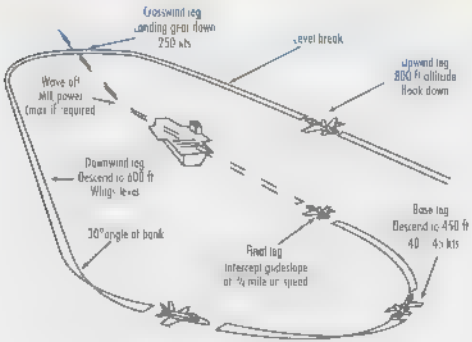
So, if the CATCC instructs you to enter the pattern on the crosswind leg, you should approach the carrier from the starboard side, passing 1 to 1½ nm in front of the bow to enter the pattern. See the carrier landing pattern diagram for altitude and speed information for each leg.

Entering the pattern from astern

The preferred approach is from astern, entering the pattern at the standard overhead recovery point (the upwind leg).

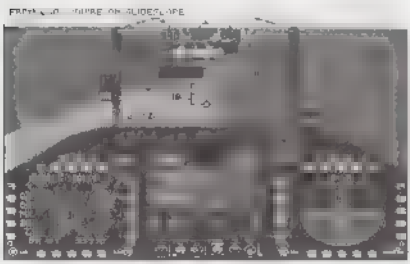
At 4 nm, you should be at 800 ft AGL, traveling at 250 kts. Drop the tailhook (press H), and level out.

You will fly over the deck of the carrier, turn 90° into the crosswind leg, then turn 90° into the downwind leg, then turn 90° into the base leg, and finally turn 180° into the final approach.



- **Upwind leg:** Pass over the carrier at 250 kts. Once you're abeam the island, roll into a bank of about 30° to 35°.
- **Crosswind leg:** Turn to fly parallel to the carrier until you reach 1½ miles abeam the ship. Lower the gear (press G). Reduce speed to approximately 180–200 kts.
- **Downwind leg:** Continue your turn to a heading opposite that of the ship's course. Descend to 600 ft. As you approach the end of the downwind leg, level off. Adjust your speed to 140 kts. Lower the hook (press H). Watch the LSO for a "Hook" message. If you see a "Wave off" message, it means that you're not just meeting the requirements for the landing signal officer (LSO) for that approach. Execute a wave-off.
- **Base leg:** When you're abeam the island, start a gentle turn toward the carrier. Look out your left window to spot the carrier. Roll left again and continue your descent. Aim for a point about ¼ mile behind the ship. Descend to 450 ft and slow to 145 kts. During the base leg, the LSO will transmit either an "Okay" or an "Adjust" message (see "LSO messages" later in this chapter).

- **Final approach** Continue turning, taking care to line up with the landing deck. You should now be about a mile from the carrier, flying at 400–450 kts, 450 ft AGL, with hook and gear down. Lined up with the carrier landing deck. Note that the centerline of the landing deck is offset 9° to port of the carrier centerline. For example, if the carrier's heading is due north (000°), the heading of the landing deck is 350°.



If you are within landing parameters, you should be receiving instructions from the LSO (see *LSO messages* later in this chapter). If you aren't receiving LSO messages, you can request landing instructions from the LSO by pressing Alt+H. Skip ahead to *Final approach and touchdown* later in this chapter.

If you're outside landing parameters, don't be a hero. If the LSO signals a wave off, abort the landing and go around again for another attempt.

Instrument (or straight-in) landings

The second carrier landing method is called an instrument landing. In an instrument landing, it's assumed that you cannot see the carrier and are, therefore, unable to enter the landing pattern. Your plane's ILS and the landing signal officer (LSO) guide you in on a straight-in approach.

If it's at night, you're low on fuel, or your plane's maneuverability is impaired, you should opt for a straight-in landing instead of a pattern landing.

At about 10 miles from the carrier, be within the following parameters:

- Within a 45° cone behind the landing deck (22.5° on either side of the centerline of the landing deck)
- Between 1,200–5,000 ft AGL
- Airspeed 300 kts or slower

Then turn on the ILS by pressing ▢ and call the carrier's landing signal officer (LSO) by pressing Alt H. Maintain your line-up, and at about six miles, the LSO will ask you if you're a straight-in approach and landing (see *LSO messages* later in this chapter). The bars on the ILS show where you are in relation to the proper glide slope and flight path.

Final approach and touchdown

Assuming you are lined up with the landing deck, one mile out, at 140–145 kts with hook and gear down, you are ready for touchdown.

Make minor adjustments with stick, rudder and throttle to get yourself on the proper alignment, glideslope and airspeed. There are several landing aids to guide you in: see *LSO messages* and *The meatball* below. If you want to use the ILS, see Lesson 1 in Chapter 5, Air Combat Training, which explains the ILS in detail.

During the final stages of landing, the flight controls are seemingly reversed. At slow airspeeds, the throttle controls altitude while the stick controls speed.

When you are so close to stall speed, pulling back on the stick will not cause you to gain altitude; what it will do is lower your airspeed and increase your rate of descent. When you increase throttle, you slow or stop your rate of descent.

As you adjust your final approach, use the throttle for altitude control and the stick for speed control. If you are going too fast, ease back on the stick. If you are too high, ease back on the throttle.

Above all, do not overcontrol. Use very small, gentle movements of the controls to adjust your approach.

As you touch down, open the throttle. If the tailhook misses the arresting wire for some reason, you'll need the power to get back off the deck. If this were a real carrier, there's nothing past the end of that short deck but saltwater and fish.

Once your tailhook catches an arresting wire, you will decelerate rapidly. Close the throttle, set the wheel brakes (press W) and fold the wings (press Ctrl W) if you're home.

"Bolters"

The term "bolter" applies to a carrier landing where the tailhook fails to catch an arresting wire. If you find yourself outside of the speed, attitude or angle of attack parameters given here, be ready to immediately give it the gas, pull up, and try again.

LSO messages

As you're trying to set that jet on what looks like a moving postage stamp, your best friend is the landing signal officer (LSO). As you turn in for a final approach, the LSO will inform you of your status with the following messages.

Transmission	Meaning
"The deck is fouled."	The deck is obstructed such that it is unsafe to land.
"The deck is clear."	The deck is clear of obstructions. It is safe to land.
"Check your lineup."	Your aircraft is not properly lined up for landing.
"Drop your gear."	Lower your landing gear (press G).

"Drop your hook."	Drop your tailhook (press H)
"Gear down, hook down, roger ball"	Your landing status is verified: your gear is down, your tailhook is down. Begin following the meatball.
"Level your wings."	Your plane is banking. Level out.
"You're on glideslope."	You are on the proper glideslope for landing.
"You're a little high."	Your aircraft is above the proper glideslope for landing.
"You're a little low."	Your aircraft is below the proper glideslope for landing.
"You're drifting left."	Your plane is to the left of centerline.
"You're drifting right."	Your plane is to the right of centerline.
"Wave-off."	Abort your landing; go around again. This is mandatory.
"Bolter!"	You missed the arresting wires. Power up immediately, lift off and go around again.
"Power!"	Your aircraft is under powered or your rate of descent is too high. Increase throttle immediately.
"Climb!"	Pull back on the stick immediately to gain altitude. You are dangerously low.

After a successful landing, the LSO will grade your landing and tell you which wire you caught with the three wire being preferred. The landing grades are:

Grade	Meaning
Perfect Pass	You maintained an optimum glideslope and lineup throughout the landing approach.
Good Landing	You had minor deviations in the landing approach, but made good corrections.
Fair	You had minor deviations in the landing approach.
No Grade	Your approach was below average (for example, you dropped your hook or lowered your gear late), but you landed safely.
Unsafe	There were gross deviations in your landing approach inside of 2,000 ft and/or you ignored the wave-off call.

Calling the LSO

If the LSO is not giving you landing instructions, it is probably because you are not far outside landing parameters that it doesn't look like you are trying to land. To request landing instructions from the LSO, press

Alt M. Remember the LSO only gives you final approach instructions. Don't bother calling the LSO until you are within reasonable landing parameters.

The meatball

The carrier is equipped with special deck lights (called the "meatball" or "ball") by pilots that show you whether you are above or below the proper glideslope. The meatball is a stack of three lights, two red and one green, with lenses that create a different optical effect depending on your glideslope.

You will not be able to see the meatball until you are within a mile of the carrier and even then it may be difficult to discern. Look just to the left of the landing strip.



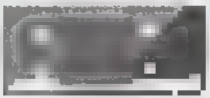
If you see the red light above the green lights, you are above the glideslope. Descend until you see only the green lights.



If you see the bottom red light below the green lights, you are below the glideslope. Ascend until you see only the green lights.



If you see only the green lights in the middle you are right on the glideslope.



The yellow lights on either side of the meatball are called the "wave off" lights. They tell you to abort your landing, possibly due to a fouled deck or other hazardous landing condition. If you see these lights, abort your landing and go around again.

Autopilot landings

The easiest way to land is to use the autopilot. Press [A] at any time to engage the autopilot, which will go a standard visual landing. If you want a straight-in autopilot landing, you need to use the automated carrier landing system (ACLS).

The ACLS can be engaged any time you are within the following parameters:

- Between 37 nm and 6 nm from the carrier
- Within a 45° cone behind the landing deck (22.5° on either side of the centerline of the landing deck)
- Between 1,200–5,000 ft AGL
- Airspeed 300 kts or slower

To engage the ACLS,

- Activate the ILS (press [ ])
- Engage the autopilot (press [A])

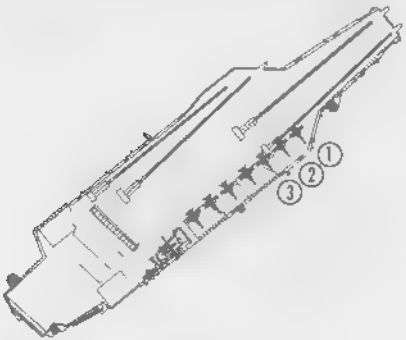
The ACLS will then guide you in for a straight-in, hands-off landing.

Wingmen and landing

If you have wingmen, you must take extra care when landing. As soon as you reach the final waypoint of your mission, your wingmen will begin turning up for

the landing sequence. They will space themselves out to give each other time to land. You have time to land before the first of your wingmen arrives, but if you take too long, the rest of your flight will arrive.

If you do land first, be sure to taxi off the landing strip to make room for your wingmen. As the first plane down, you should taxi to the first open space. The following illustration shows the order in which aircraft should be parked.



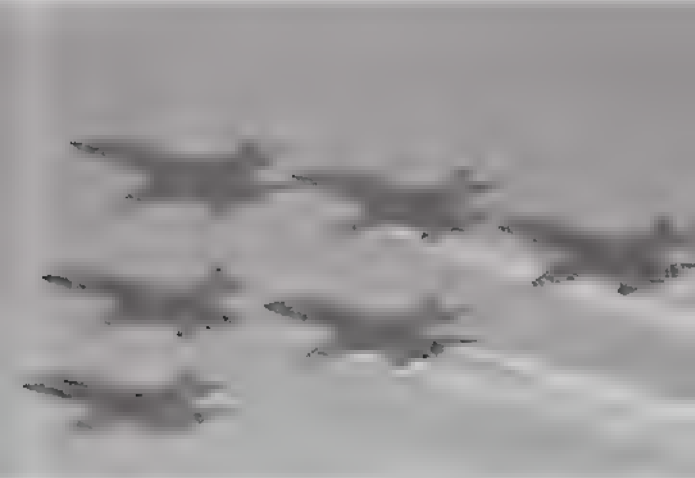
The LSO and multiple aircraft

The LSO only gives course correction messages to the aircraft that is on final approach. If you are not on the final approach leg and you hear course corrections on the radio, ignore them; they are directed towards your wingman.

Once the rest of your flight has entered the landing pattern, it may be best to circle awhile as they land. Then, once they are all safely on deck, you can land yourself.

CHAPTER
7

COMMUNICATIONS



You can play *Hornet* with human allies or enemies by linking your computers in "head to head" mode. Two people can fly together or against each other using a direct connection, modem or Novell NetWare compatible network. Two to six people can play over a Novell Netware network.

The basic communication features of *Hornet* are the same as those of *Falcon 3.0* or *MiG 29*. For information on configuring your modem or network, dialing the phone and setting communication parameters, see your *Falcon 3.0* or *MiG 29* manual.

When you play in head to head mode, each computer must have a registered copy of one of the following programs:

- *Hornet*
- *MiG 29* version 1.02
- *Falcon 3.0* version 3.03

All of these programs can communicate with one another. For the purposes of this manual, we assume that at least one player is running *Hornet*.

Falcon 3.0 and *MiG-29* compatibility

Since *Hornet* is only communications-compatible with version 3.03 of *Falcon 3.0* and version 1.02 of *MiG 29*, both *Falcon 3.0* and *MiG 29* must first be updated to the latest versions before beginning head to head games.

When you install *Hornet* to your FALCON3 or MIG29 directory, your copies of *Falcon 3.0* and *MiG 29* are automatically updated to version 3.03 and 1.02, respectively.

An updater has been provided on Disks 1 and 2 to update *Falcon 3.0* and *MiG 29* for communications compatibility. You may freely distribute copies of Disks 1 and 2 to other *Falcon 3.0* and *MiG 29* owners for this purpose. Please do not distribute copies of Disk 3.

Updating *Falcon 3.0* and *MiG-29*

Foolow the instructions below to update your copy of *Falcon 3.0* or *MiG 29*

1. Insert *Hornet* Disk 1 into a floppy drive
2. If you put the disk in drive A, type **A:** and press Enter. If you put the disk in drive B, type **B:** and press Enter.
3. At the A or B prompt, type **SHUPDATE** and press Enter.
4. On the screen that appears, you will be given the choice to "Continue..." or "Quit." Select "Continue" to go on.
5. A warning message appears informing you that a your old squadrons will be deleted if you continue the update.
6. The update program now searches your hard drive for *Falcon 3.0* or *MiG 29*. When it finds either game, the updater will say, "Falcon 3.0 found in <directory name>" or "MiG 29 found in <directory name>." You can edit this path if it is incorrect.
7. At the bottom of this screen are four buttons: "Install," "Search," "View" and "Quit." "Quit" aborts the update. "View" brings up the Read Me file. "Search" continues the search for other copies of *Falcon 3.0* and *MiG 29*. "Install" continues the update.
8. Select "View" and read the Read Me file which contains last minute information. Then select "Install."
9. The update program will begin to update your copy of *Falcon 3.0* to version 3.03 and your copy of *MiG 29* to version 1.02.
10. Partway through, your computer will beep and ask you to insert Disk 2.
11. When the updater is finished, it will beep. Press any key to return to DOS.

Type **CHECKVER** from your FALCON3 or MIG29 directory to confirm that your copy of *Falcon 3.0* or *MiG-29* has been successfully updated.

The Caller

In any head-to-head game, one player must take charge of the communications process. This player—the Caller—should have the fastest computer.

If any of the players are using *Falcon 3.0*, one of them must be the Caller. Otherwise, the Caller must be one of the players running *Hornet*. This is the only difference between the two programs in a communication session. To the Caller, *Falcon 3.0* players appear identical to *Hornet* players.

Head-to-head game formats

There are many ways to join with other players in head-to-head games. The two basic scenarios for head-to-head play are:

- Duel, where all players are adversaries
- Campaigns, where players cooperate against the enemy

Both scenarios accommodate all three game programs simultaneously. For instance, in a six-player campaign, two of the players might run *Hornet*, two *Falcon 3.0* and two *MiG-29*. This situation is a *multigame* scenario. Naturally, *single game* scenarios in which all the players use one game program, work fine.

When you combine the possibilities for games and scenarios, you come up with these types of head-to-head encounters:

- Single-game duels
- Multigame duels
- Single-game campaigns
- Multigame campaigns

The final variable is the number of players. Each situation in the above list can accommodate two to six players.

Duels

A duel is a free-for-all in which one player emerges victorious. The game continues in this way forever or until only one player remains or until the players quit.

In the default *Falcon 3.0* and *Hornet* players carry four AMRAAMs, two AIM-9Ms, two AIM-9Ps and an ECM pod. All MiG-29 players have two R-27Rs, two R-73s and two ECMs. The Caller turns off the ammo, AMRAAM setting on the Realism Flight Levels screen. All players have every available weapon in unlimited quantities.

In a multiplayer duel, the IFF indicator recognizes which planes are friends and which are enemies. If you're a Hornet pilot, your IFF recognizes Falcons and other Hornets as friendly and MiG-29s as enemies. MiGs appear friendly to other MiGs with Falcons and Hornets as the enemy.

Single-game duels

For all players in a duel to have *Falcon 3.0* all of them have one of the other two games. It's possible to set up a duel. One player, the Caller, initiates the communication session.

If there are just two players, both pilots begin with both players' names in ACCEPT or SAVE SETTINGS on the Communications Setup screen.

In a multiplayer duel, the game begins when all the players have accepted the invitation from the Caller. The duel ends when there is only one survivor or when the Caller ends the duel by pressing Alt-Q (to start a new duel) or Alt-X (to exit to DOS).

Multigame duels

When players use different game programs, there are some special considerations for setting up a head-to-head encounter.

The main rule to remember is the one about the Caller's game. See *The Caller* above for a recap of this rule. Here are the specifics for a multigame duel:

Falcon 3.0, that player must be the Caller. Otherwise, the Caller must have *Hornet*. This rule holds for multigame duels regardless of the number of players.

If there are any *MiG-29* players in the crowd, the protocol becomes a bit more complex. When MiGs are present, the software divides the players into sides with Falcons and Hornets opposing MiGs. This division is crucial when playing a campaign in head-to-head mode. Even though there aren't any "sides" in a duel, the protocol applies when you set up the communication session.

There is an additional rule of protocol when MiGs are involved. The Caller must invite all Falcon pilots first, then Hornet pilots, followed by the MiG players.

Furthermore, if two or more players have *MiG-29*, one of them must act as the "Master MiG." The Master MiG's computer is a communications hub for the other MiG players. For this reason, the Master MiG should be the MiG player with the fastest computer.

The next section shows how to combine all of these variables when you set up a head-to-head session with a mixture of Falcons, Hornets and MiGs.

Setting up a multiplayer/multigame duel

Voice communication is very important in a multiplayer/multigame duel. The players should be within earshot or available by phone during the setup period.

First, each player must decide which plane to fly (that is, which game program to run), and let the others know. When the players agree on who flies what, the next step is to choose a theater. Everybody must agree on this choice, too.

Finally, the players select a Caller and a Master MiG. It is the Caller's responsibility to make sure that the rest of the players join the game in the correct order. The sequence *must* be as follows:

1. Caller: a *Falcon 3.0* player if there is one.
2. All other *Hornet* and/or *Falcon 3.0* players.
3. Master MiG.

4. All other *MiG-29* players

Once all the preliminary details are settled, and the Caller knows when to invite each player, the communication process begins. It works like this:

1. Before connecting, all players select the same settings for Connection, Game Type and Baud Rate.
2. The Caller sets the Mode switch to CALL. All other players select ANSWER.
3. The Caller clicks ACCEPT.
4. All *Falcon 3.0* and *Hornet* players choose ACCEPT or SAVE SETTINGS.
5. The Caller sees a box with the network names of all the other players. The Caller invites another player to join by choosing a name and pressing OK. The Caller must invite only *Falcon 3.0* players first, and then *Hornet* players second.
6. After all the *Falcon 3.0* and *Hornet* players have accepted, the Master MiG should then choose ACCEPT or SAVE SETTINGS.
7. The Caller invites the Master MiG.
8. Once the Master MiG has accepted, the other *MiG-29* players select ACCEPT or SAVE SETTINGS.
9. The Caller then invites any remaining *MiG-29* players.
10. When all players have accepted, the Caller selects EXIT to begin play. In a six player game, play starts immediately after the sixth player accepts.

The *MiG-29* flight leader should click the TAKEOFF button first, followed by the *Falcon 3.0* or *Hornet* leader within 10 to 20 seconds.

Campaigns

In a head-to-head campaign, the players fly missions in a theater of war. Unlike dueling, players cooperate against the enemy. If all the players fly on the same side (for instance, if they all have *Falcon 3.0* or *Hornet*), the computer controls the enemy. In a game involving Falcons, Hornets and MiGs, the players take sides.

Each side in a head-to-head campaign has a flight leader. In single-game campaigns, the Caller is the flight leader. In multigame campaigns, the Caller is the Falcon, Hornet flight leader and the Master Mig is the MiG flight leader.

The process of setting up a head-to-head campaign is like the procedure for a duel. The Caller must name the players to join in the order shown in the last section.

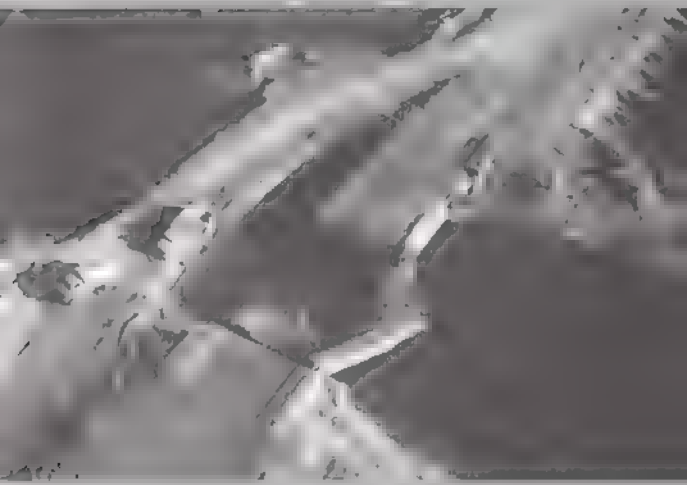
Once all the players are in the game, a series of campaign news screens appears. Then the players get their instructions for the mission.

At this point, the other players on each side go back and watch as the flight leader plans the mission. He or she sets the waypoints, arranges the flights and wingmen and finally assigns weapons. When the flight leader clicks TAKEOFF, the action begins.

The mission in a multigame campaign pairs both groups of players into contact. For example, all the Falcon/Hornet pilots are assigned to protect a bridge. The MiG pilots might be assigned to fly CAP (combat air patrol) over the same bridge.

CHAPTER
8

THEATER BRIEFING



"Unfortunately, we are not normal people"

Bosnian Serb leader Radovan Karadzic



Bosnia-Hercegovina: The enigma and the terror

The dream of a unified Yugoslavia is over. Idealism couldn't hold it together; neither could Communist dictatorship.

The "land of the South Slavs" began to unravel in 1991 with the secession of two territories, Croatia and Slovenia. A third territory, Bosnia-Hercegovina, followed a year later and plunged immediately into civil war.

Today, Bosnia, as it's known for short, is sliding into the final stages of self-destruction. The war is driven by ancient hatreds that outsiders strain to understand. Ethnic groups who kept an uneasy peace under Communism are settling old scores with a savagery that has led many to comparisons with the Holocaust.

As Bosnia unravels, the world watches in disbelief and frustration. Liberals and conservatives, facing tragedies comparable even in a huge country, are again in agreement. Can the Western powers do anything to stop further bloodshed? If history is the guide for Bosnia, is there any way to prevent similar tragedies in Kosova and Macedonia? And if there are any answers, they begin in the past.

Historical roots

The Balkan region has always been one of history's fault lines. In the fourth century, when the Roman Empire split into eastern and western halves, the boundary ran down the middle of the area now known as Bosnia-Hercegovina. When Rome fell, the region changed from an imperial boundary into a religious one.

In 1052, the first great schism of Christianity separated Roman Catholicism in the west from the Orthodox in the east. A large part of the Bosnian population rejected both forms of Christianity in favor of the Gnostic teachings of the Bogomil sect. When the Ottoman Turks conquered the region in 1463, many of these Bosnians converted to Islam.

The religious divisions remain today. Of the inhabitants of the former Yugoslavia's largest republic, primarily Serbs are Eastern Orthodox, 32% are Roman Catholic, with the Croats making up the majority, and 13% are Muslim. The Bosnian Muslims, whose culture is similar in many ways to their Serb and Croat neighbors, have suffered the most in the current tragedy.

Bosnia in the World Wars

At the birth of the modern world, Bosnia was again caught between empires. The decline of Ottoman power provided an opening for the Austrian Empire, which annexed Bosnia in 1908. Bosnian Serbs chafed under Austrian rule. When a Serb assassin killed Austrian Archduke Franz Ferdinand in Sarajevo, the capital of Bosnia, the result was World War I.

After the war Bosnia was incorporated into the Kingdom of Serbs Croats and Slovenes the forerunner of modern Yugoslavia. This Serb dominated monarchy was unstable and unpopular from the beginning and was attacked in turn when the forces of Hitler and Mussolini arrived in 1941.

In the Balkans the war unleashed ethnic violence that rivaled the excesses of the Nazis. Hitler's Croat puppets the Ustashi carried out a campaign of genocide against the Serbs. And organized Serb partisans the Chetniks responded with equal brutality. By the end of the war two million Yugoslavs were dead. Half of them at Yugoslav hands.

Outsiders who wish to understand the current crisis must begin with World War II. The memory of genocide explains the current bitterness. It is a memory that cuts both ways. All of the parties in the current tragedy have been both aggressor and victim.

Tito's Yugoslavia...and afterwards

The leader of the anti fascist Partisan Resistance Movement was a Croat strategist named Josip Broz Tito. His code name in the Comintern was Croat Communist Party.

When the Nazis left Yugoslavia in 1944 Communist partisans took power with great popular support. Their leader Tito set up a Marxist state with a difference. For a post war Communist country Tito's Yugoslavia was remarkably independent. Having fought a war of national liberation Tito kept Yugoslavia free of Soviet domination while maintaining ties with the Western democracies. He performed a similar balancing act at home playing one ethnic group off another for decades. Tito's personal power combined with the fear of a Soviet invasion held Yugoslavia's inner demons at bay.

In 1974 Tito became "President for Life" and died six years later. His Marxist government tottered on until Communism itself fell apart.

Democracy comes to the Balkans

In 1990 the republics of Yugoslavia held democratic elections. Bosnia, Croatia, Slovenia and Macedonia elected non-Communist governments. These new governments placed national and ethnic interests above any ideal of Yugoslav unity.

On the other hand, Serbia's repackaged Communists, led by President Slobodan Milosevic, wanted to keep the Yugoslav federation intact. Their primary motive was fear of the central authority collapsed, what would happen to the 25% of Yugoslavia's Serbs who live outside Serbia? Memory runs deep in the Balkans.

Croatia and Slovenia declared unilateral independence in June 1991. War began immediately. When it ended in January 1992 one-third of Croat territory was in Serb hands. Bosnia declared independence a month later.

The resulting conflagration of Serbs, Croats and Muslims horrified the world. It is a war of vengeance directed largely at civilian targets. Snipers and artillery have decimated Sarajevo and other cities. Of the 24 million people in prewar Yugoslavia, 3.5 million are now refugees.

Campaign rationale

It is the year 1994. After repeated and unsuccessful appeals to the Western powers, the Bosnian Muslims receive military and financial aid from fellow Muslims in Turkey. The specter of Turkish influence in Bosnia provokes an immediate reaction from Turkey's ancient rival, Greece. While the Greeks threaten to intervene on behalf of the Serbs, the conflict seems poised to escalate into global war.

Finally, the United Nations intervenes and sends peacekeeping troops to Sarajevo and other besieged areas in Bosnia. For a moment, the world holds its breath.

Meanwhile, Serbia appeals for international aid to assist its scattered kin in Bosnia and Croatia. Russia responds by sending dozens of MiG-29s to Serbia for hard currency. The Bosnian Serbs, encouraged by the apparent

support of Greece and Russia, move in for the final kill.

As the pressures on the increasingly lawless Serbs become increasingly gruesome and apocalyptic, NATO commits several Marine units and a carrier task force, led by the USS Theodore Roosevelt (CVN-71), to Bosnia-Herzegovina. Its objective is to establish a corridor that allows the UN peacekeeping forces to go on the offensive and become *peacemakers*.

The goal of your campaign is to force the Serbs and Croats to accept a cease fire and come to the bargaining table. Your strategy is to drive them back to their original positions, and further if possible.

In a Hornet only campaign you will be based on the USS Roosevelt in the Adriatic Sea. In a combined arms campaign where you fly alongside Air Force F-16s (see Chapter 7 'Communications') you will be based in Sarajevo.

Campaign objectives

The only hope for a lasting resolution to this crisis is if Serbia can be made to accept a cease fire and come to the negotiating table. To do this the UN forces must first break the Serbian choke hold on central Bosnia and then drive into Serbia itself, with the Serbs agree to stop fighting.

The key points in the UN battle plan are mobility, self defense and protection of civilians. To meet these goals your squadron will have to accomplish the following tasks:

- Protect Sarajevo and other major urban centers from Serbian shelling and bombing. The mountains around Sarajevo are especially important as artillery in those positions can rain a heavy barrage on many civilian areas and UN bases.
- Keep the roads and bridges clear and under UN control. Serbian troops that take control of these vital transportation arteries will have to be forcibly removed.

- Protect air friendly air bases. The Serbian forces have the advantage of numbers and air defenses. The UN's only edge is its air superiority. U.S. air power is crucial to the UN battle plan so keep those runways open.
- When the advance into Serbian territory begins, recon squadrons will be tasked with Wild Weasel (SEAD) and bombing missions to eliminate the Serbian air defenses and infrastructure. If you can penetrate the complex Russian style Serbian air defense network and destroy its supply and communications, the Serbians will realize the futility of their position and will be much more likely to agree to a cease fire.

In all cases, precision weapons delivery is critical. Given the grievous harm already done to the infrastructure and population centers of Bosnia Herzegovina, further damage from stray ordnance could be catastrophic.

APPENDICES



Appendix A

Technical Specifications

F/A-18 Hornet



Type:	Carrier- and land-based strike/fighter
Length:	56 ft 0 in
Wingspan:	37 ft 6 in
Wing area:	400 sq ft
Takeoff weight:	51 900 lb (attack mission) 36,710 lb (fighter mission)
Max weight:	56,000 lb
Max payload:	15,500 lb
Engines:	Two General Electric F404 GE-402 EPE turbofans each rated at 17,600 lb static thrust with afterburner
Top speed:	Mach 1.8 level flight, 920 knots at 35,000 ft
Max climb:	35,000 ft/1 5 min
Ceiling:	50,000 ft
Corner velocity:	400 kts (sustained)
Turn radius:	1,592 ft (instantaneous)
Turn rate:	25 deg/sec (instantaneous) 18 deg/sec (sustained,
Max range:	1,800 nm
Combat radius:	290 nm
Max fuel:	10 860 lb (internal) 6,732 lb (externa.)
Flight controls:	Quadruple redundant, ful digita. fly-by-wire
Avionics:	AN APG-65 multi mode digital air to-air, air-to-ground tracking

radar. AN/ALR RWR. AN/ALQ 126B ECM. FID 2035 horizontal situation display. AN/ALQ 165 Jammer. Two AN/AYK-14 digital computers. AN/ASN 130A inertial navigation system. AN/AVQ-28 HUD.

USS Theodore Roosevelt CVN-71



Builder:	Newport News Shipbuilding
Laid:	31 October 1981
Launched:	27 October 1984
Commissioned:	25 October 1986
Displacement:	77,606 tons light 97,574 tons full load
Payload:	2,970 tons of aviation ordnance, 3.5 million gallons of aircraft fuel
Length:	1,040 ft waterline 1,092 ft overall
Beam:	134 ft
Flight deck:	250⅚ ft
Angled deck:	9 degrees 3 ft canted to port, 796⅔ ft long
Hangar deck:	684 ft long, 108 ft wide, 26½ ft high
Draft:	38½ ft
Propulsion:	4 steam turbines, 280,000 shp; 4 shafts
Reactors:	2 pressurized water A4W
Fuel cores:	Service life of 23 years, providing nearly 2 million nm steaming
Speed:	30+ knots
Manning:	3,162 2,020 Air Wing
Aircraft:	24 F-14 Tomcats, 24 F/A-18 Hornets, 10 A-6E Intruders; 4 KA-6D Intruders;

	4 E-2C Hawkeyes, 4 EA-6B Prowlers, 10 S-3A Vikings, 6 HH-60H Seahawks.
Catapults:	4 steam Mk 3-1
Elevators:	4 deck edge; 85 x 52 feet, 130,000 lbs capacity
Missiles:	3 Mk 29 8-tube NATO Sea Sparrow launchers
Guns:	4 Mk 16 20 mm Phalanx CIWS
Radars:	SPS-48 3-D air search, SPS-49 air search, SPS-64(V) navigation, SPS-67(V) surface search, SPN-42 Automatic Carrier Landing System
Fire control:	3 Mk 91 missile FCS
EW systems:	SLQ-29, SLQ-17 + WLR-8

The Theodore Roosevelt is the fourth ship of the Nimitz class of aircraft carriers. The contract for the construction of the ship was awarded to Newport News Shipbuilding on September 30, 1981. Construction began on October 8, 1984, and less than a year later, President Reagan signed an Executive Memorandum naming the CVN-71 after the 26th President of the United States. USS Theodore Roosevelt was commissioned and placed into active Navy service on October 25, 1986.

Roosevelt has a combat displacement of 97,000 tons and is 1,040 feet in length at the waterline. The area of the flight deck is 4,092 ft long by 252 ft wide. Two aircraft catapults are located forward and two on the angled deck aft of the ships. Using all four catapults, it is possible to launch aircraft one every 20 seconds.

USS Theodore Roosevelt spent almost half of 1988 under way, most of it being for two- and three-week carrier qualifications and operational training exercises. In early March, TR operated with all of the squadrons of Carrier Air Wing Eight (CVW-8), comprising 85 tactical aircraft for the first time. On December 30, 1988, Roosevelt was underway for her first operational deployment to the Mediterranean Sea.

On December 28, 1990, Theodore Roosevelt left Norfolk, Virginia for her second Mediterranean Deployment and for Operation Desert Storm. TR entered the air war against Iraq on January 19, 1991. CVW-8 flew an unprecedented total of 4,209 sorties, more than any other carrier in the war. TR strike aircraft dropped 4,843,233 pounds of ordnance with uncanny accuracy, until a cease-fire was ordered on February 28th. Roosevelt completed her six month deployment and was relieved by the USS Forrestal Battle Group on June 28, 1991.

Theodore Roosevelt sailed on her third Mediterranean Deployment in March 1993. The U.N. Security Council passed Resolution 836 to enforce the sanctioned air exclusion zone (AEZ) over Bosnia and Secretary of Defense Les Aspin approved an order directing U.S. aircraft to join NATO's support of the U.N. Protection Force in Bosnia. In response to the order, USS Theodore Roosevelt transited the Suez Canal to begin Operation Deny Flight in the Adriatic Sea. The carrier launched about 1,000 sorties over Bosnia. This represents 12-18 sorties out of a total of 80–90 sorties launched daily.

Appendix B

Naval Medals

The following medals may be awarded to Hornet pilots. They are listed in order of precedence except for the Purple Heart, which is described at the end.



Navy Medal of Honor

The Navy Medal of Honor is the highest award that the Navy or Marines can bestow. It is given in the name of Congress to persons who distinguish themselves by extreme bravery risking their lives above and beyond the call of duty while actively combating an enemy of the United States or its allies.

The Medal of Honor is a very rare award. It is only given in extraordinary circumstances.

Navy Cross

The Navy Cross is awarded for tremendous heroics that don't quite justify a Medal of Honor. It is given to any person serving with the U.S. Navy or Marines who is engaged in action while a member of the United States or who serves with friendly foreign armed forces engaging opposing foreign forces.

The primary difference between this award and the Medal of Honor is the degree of heroism and actions taken

Silver Star

The Silver Star is awarded to a member of any branch of the U.S. Armed Forces for gallantry when engaged in action with an enemy of the United States or when serving in conjunction with allied forces engaging opposing foreign forces.

Distinguished Flying Cross

The Distinguished Flying Cross is awarded to a member of any branch of the U.S. Armed Forces who has distinguished herself by heroism or extraordinary achievement while participating in aerial flight. The requirements for this award need not involve the engagement of enemy forces.

Bronze Star

The Bronze Star is awarded to members of any branch of the U.S. Armed Forces who have distinguished themselves by heroism or meritorious achievement or service, not involving participation in aerial flight, while engaged in action with an enemy of the United States or when serving in conjunction with armed forces engaging opposing foreign forces.

Air Medal

The Air Medal is awarded to members of any branch of the U.S. Armed Forces who have distinguished themselves by meritorious achievement while in aerial flight. This award is given for both combat and non-combat actions.

Purple Heart

The Purple Heart is awarded to any member of the U.S. Armed Forces who is wounded while in combat. It is also given posthumously to the next of kin for those who died from their wounds or were killed in action.

Earning medals

A pilot may receive any medal after only one sortie, but it is more likely that a medal will be awarded after repeated successful flights. Pilots earn recognition for such things as shooting down enemy fighters or bombers, assisting in an air-to-air kill, destroying or damaging vital primary targets, and for completing your mission.

Appendix C

Glossary

ACLS – Automatic Carrier Landing System. The system for automatic landing of aircraft onto the carrier's deck. Steering commands are sent to the aircraft via data link for a total hands off landing.

ACLS window – an area in space normally about six miles from touchdown in which an aircraft is acquired by radar for ACLS control.

ACM – Air Combat Mode radar mode.

ADI – attitude display indicator.

Air Wing – the complete complement of aircraft on board an aircraft carrier.

Amidships – in the middle of a ship.

Angle of attack – angle of the wing to the flight path of the aircraft, determines amount of wing lift at a given speed.

ASL – azimuth steering line.

Aspect angle – angle created by the intersection of your line of sight to the target and the target's velocity vector.

Astern – behind a ship.

Ball – the primary optical landing aid, also known as FLOLS.

Barometric altitude – the altitude measured by air density.

Base recovery course – the ship's magnetic heading during flight operations.

Beam – the widest point of a ship's hull.

Bolter – an unsuccessful landing in which the tailhook fails to catch an arresting cable.

Bow – the front of a ship.

Brown shirt - the plane captain

Carrier Air Traffic Control Center (CATCC) – the centralized agency responsible for the status keeping of all carrier air operations and control of all airborne aircraft.

Carrier control area – a circular airspace around the ship and under the cognizance of CATCC.

Cat (catapult) – steam powered system used to accelerate aircraft to flying speed during launch.

CIC – Combat Information Center.

COMED – combined map electronic display.

Cross-deck pendant – the portion of the arresting cable which stretches across the deck and engages the hook.

CVW – abbreviation for Carrier Air Wing. *See Air Wing.*

Cyclic Ops – alternating launches and recoveries.

DDI – digital display indicator.

DIL – displayed impact line.

Displacement – the amount of water (in tons) that a ship's hull replaces in any body of water.

Dropline – a row of orange lights extending vertically down the stern from the centerline, which enables the pilot to make adjustments in lineup on final approach.

Final bearing – the magnetic bearing assigned by CATCC for final approach. It is an extension of the landing area centerline.

FLOLS – Fresnel Lens Optical Landing System. The primary optical landing aid, called the "ball."

Fouled deck – any condition in which the carrier deck is not ready to receive aircraft.

Glide path the path of descent to landing

Glideslope an electronic or optical beam marking the glide path.

G limiter the system that sets an absolute G limit on maneuvers.

Green shirt a deckhand whose duties involve readying aircraft for launch, securing them to the catapult etc

HI horizontal indicator

Hook the arresting hook which is lowered from the tail of the aircraft for landing approach and which catches the arresting cable.

HSI horizontal situation indicator

Island the portion of the ships superstructure above the flight deck.

Knot abbreviation for nautical mile or 1.15 statute miles.

LSO Landing Signal Officer, responsible for all aircraft recoveries. The LSO judges the quality of each approach, guides and critiques pilots on landings

Meatball - See *Ball*

MFD multifunction display.

NATOPS Naval Air Training and Operating Procedures Standardization. Every naval aircraft has a NATOPS manual which explains the aircraft systems and operating procedures.

Pickle switch a switch by which the LSO terminates the wave-off lights to abort a landing attempt

Port nautical term for left

Purple shirt a deckhand in charge of fueling aircraft

Radar altitude the altitude measured by a radar beam

Red shirt a deckhand that handles ordnance

RWR – radar warning receiver

SAD – Situation Awareness Display.

SAM – Situation Awareness Mode.

SCN – Scan radar mode.

SEAD – suppression of enemy air defenses.

Spot the deck – to look at the deck rather than the ball during landing.

Starboard – nautical term for right

Stern – the back of a ship.

STT – Single Target Track radar mode.

TACTS – Tactical Air Combat Training System

Tailhook – See *Hook.*

Trap – a successful, arrested landing aboard an aircraft carrier

TWS – Track While Scan radar mode.

Vultures Row – an observation deck on the island from which flight operations can be seen.

Wave-off – a signal from LSO to pilot aborting a landing. Compliance by pilot is mandatory

White shirt – a deckhand responsible for inspecting aircraft about to be launched.

Yellow shirt – the director of movement of aircraft on the flight deck.

Appendix D

Bibliography

F/A-18 Hornet in Action. Scott Lindberg. Blue Ridge Summit, PA: Tab Books, Inc., 1982.

F/A-18 Hornet. Mike Wanat. Warplane Classics. Shaw Osceola, WI: Motorbooks International, 1987.

F/A-18 Hornet. Roy Braybrook. New York, NY: Crescent Books, 1991.

F/A-18 Hornet. Bill Gunston. Osceola, WI: Motorbooks International, 1985.

Fighter Aircraft Tactics and Maneuvering. Robert Shaw. Annapolis, MD: U.S. Naval Institute, 1985.

Grumman Aircraft since 1929. Peter M. Bowers. London, UK: Aerospace Publishing Ltd., 1991.

Hornet. Robert F. Dorr. London, UK: Prentice Hall. *World Air Power Journal*. Aerospace Publishing Ltd. Vol. 1 Spring 1990.

McDonnell Douglas F/A-18 Hornet. Mike Spick. New York, NY: Smithmark Publishers, Inc., 1991.

Modern Air Combat. Bill Gunston & Mike Spick. New York, NY: Crescent Books, 1983.

Naval Aviation in the Pacific. John Wilson English. London, UK: Osprey Publishing Ltd., 1992.

Scramble! Thomas Halberstadt. Chattanooga Park, CA: Air Combat, June 1993.

Ships and Aircraft of the U.S. Fleet. Norman Friedman. Annapolis, MD: U.S. Naval Institute, 1990.

Strike Warfare. U.S. Naval Strike Warfare Center. Boss & Hall. Osceola, WI: Motorbooks International, 1992.

Top Gun: Modern Air Warfare. Dana Bracken & Dennis Gardner. London, UK: Osprey Publishing Ltd., 1992.

Up Check: The Naval Aviator's Complete Source Book. George Hall. Novato, CA: Presidio Press, 1987.

U.S. Aircraft Carriers: An Illustrated Design History. Norman Friedman. Annapolis, MD: U.S. Naval Institute, 1991.

U.S. Navy & Marine Corps Air Power. Jane's Publishing Co. Ltd. Editors David Donald & Jon Lake. London, UK: Aerospace Publishing Ltd., 1991.

U.S. Unique Unit Awards. NAVPERS 15,790. Washington, D.C. *Booklet*. Ship's Public Affairs Office, 1992.

The *Hornet* Team



Front Row (L to R): Kurt Boutin, Gabe Turk, Robert Cieri, Dan Deremiah, Heather Mace, Kuswara Pranawahadi, Jack Gibbons

Back Row (L to R): Russel Reiss, Eric Gratke, Steve Blankenship, Lawrence Kevin, Jim Voytilla, Andy Edlen, Michael Mancuso, Anthony Chiang, Gary Stottlemyer, Paul Mogg, J. White, Kyle Brink, Derwin Wyatt, Robert Giedt, Evan Birkby

Spectrum HoloByte, Inc. Ninety-Day Limited Warranty

To the original purchaser only, Spectrum HoloByte warrants the magnetic diskette on which this software product is recorded to be free from defects in materials and faulty workmanship under normal use for a period of ninety days from the date of purchase. If during this ninety-day period the diskette should become defective, it may be returned to Spectrum HoloByte for a replacement without charge, provided you have previously sent in your Warranty Registration Card to Spectrum HoloByte or send proof of purchase of the program.

Your sole and exclusive remedy in the event of a defect is expressly limited to replacement of the diskette as provided above. If failure of a diskette has resulted from accident, abuse or neglect, Spectrum HoloByte shall have no responsibility to replace the diskette under terms of this limited warranty.

If the diskette should fail after the original ninety-day limited warranty period has expired, you may return the diskette to Spectrum HoloByte at the address noted below, accompanied by a check or money order for the applicable replacement fee (please contact Customer Support for the exact cost), a brief statement describing the defect, and your return address. Spectrum HoloByte will replace the diskette provided that you have previously returned your Warranty Registration Card to Spectrum HoloByte, and the diskette retains the original product label.

ANY IMPLIED WARRANTIES RELATING TO THE DISKETTE, INCLUDING ANY IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, ARE LIMITED TO A PERIOD OF NINETY DAYS FROM DATE OF PURCHASE. PUBLISHER SHALL NOT BE LIABLE FOR INDIRECT, SPECIAL, OR CONSEQUENTIAL DAMAGES RESULTING FROM THE USE OF THIS PRODUCT. SOME STATES DO NOT ALLOW THE EXCLUSION OR LIMITATION OF INCIDENTAL OR CONSEQUENTIAL DAMAGES, SO THE ABOVE LIMITATIONS MIGHT NOT APPLY TO YOU. THIS WARRANTY GIVES YOU SPECIFIC LEGAL RIGHTS, AND YOU MAY ALSO HAVE OTHER RIGHTS WHICH VARY FROM STATE TO STATE.

Spectrum HoloByte, Inc.
2490 Mariner Square Loop
Alameda, CA 94501

SPECTRUM HOLOBYTE, INC. NINETY-DAY LIMITED WARRANTY

To the original purchaser only, Spectrum HoloByte warrants the magnetic diskette on which this software product is recorded to be free from defects in materials and faulty workmanship under normal use for a period of ninety days from the date of purchase. If during this ninety-day period the diskette should become defective, it may be returned to Spectrum HoloByte for a replacement without charge, provided you have previously sent in your Warranty Registration Card to Spectrum HoloByte or send proof of purchase of the program.

Your sole and exclusive remedy in the event of a defect is expressly limited to replacement of the diskette as provided above. If failure of a diskette has resulted from accident, abuse or neglect, Spectrum HoloByte shall have no responsibility to replace the diskette under terms of this limited warranty.

If the diskette should fail after the original ninety-day limited warranty period has expired, you may return the diskette to Spectrum HoloByte at the address noted below, accompanied by a check or money order for the applicable replacement fee (please contact Customer Support for the exact cost), a brief statement describing the defect, and your return address. Spectrum HoloByte will replace the diskette provided that you have previously returned your Warranty Registration Card to Spectrum HoloByte, and the diskette retains the original product label.

ANY IMPLIED WARRANTIES RELATING TO THE DISKETTE, INCLUDING ANY IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, ARE LIMITED TO A PERIOD OF NINETY DAYS FROM DATE OF PURCHASE. PUBLISHER SHALL NOT BE LIABLE FOR INDIRECT, SPECIAL, OR CONSEQUENTIAL DAMAGES RESULTING FROM THE USE OF THIS PRODUCT. SOME STATES DO NOT ALLOW THE EXCLUSION OR LIMITATION OF INCIDENTAL OR CONSEQUENTIAL DAMAGES, SO THE ABOVE LIMITATIONS MIGHT NOT APPLY TO YOU. THIS WARRANTY GIVES YOU SPECIFIC LEGAL RIGHTS, AND YOU MAY ALSO HAVE OTHER RIGHTS WHICH VARY FROM STATE TO STATE.

Spectrum HoloByte, Inc.
2490 Mariner Square Loop
Alameda, CA 94501